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Nonlinear $\mathcal{H}_\infty$ Inverse-Optimal Impedance Control for Robotic Systems

Geonho Park, Jonghoon Park, and Youngjin Choi, *Senior Member, IEEE*

***Abstract*—Impedance control is a widely adopted strategy for robotic systems physically interacting with their environments. However, it is well known that impedance control is vulnerable to disturbances and parametric uncertainties. This paper proposes a control design to impart robustness against disturbances and parametric uncertainties to impedance control, in the sense of nonlinear $\mathcal{H}_\infty$ optimality—namely, nonlinear $\mathcal{H}_\infty$ impedance control. An analytic solution to the nonlinear $\mathcal{H}_\infty$ inverse-optimal control problem is derived for the interconnected system of the robot and its environment. The resulting optimal control takes the form of reference error feedback. Robust stability is guaranteed by extended disturbance input-to-state stability. Passivity of the interconnected robot–environment system is ensured, thereby realizing robust and passive impedance control. Experimental validations demonstrate that the proposed $\mathcal{H}_\infty$ impedance control achieves robust stability and performance in the presence of disturbances and parametric uncertainties.**

***Index Terms*—Compliance and impedance control, force control, nonlinear $\mathcal{H}_\infty$ optimal control, robust control.**

I. Introduction

MODERN robotic systems are increasingly deployed in unstructured environments where they physically interact with humans and uncertain surroundings. Consequently, the ability to safely and reliably interact with the environment has become a fundamental requirement for robotic systems. Impedance control has emerged as one of the most widely adopted frameworks for achieving compliant physical interaction. Rather than controlling either motion or force, impedance control regulates the dynamic relationship between them, providing a unified framework for both free motion and physical interaction [1]–[3].

However, it is well known that impedance control is vulnerable to disturbances and parametric uncertainties. Considerable research efforts have been devoted to enhancing its robustness by incorporating robust control techniques, such as disturbance observers (DOBs) [4] and sliding mode control (SMC) [5], into impedance control [6]–[8].

In this paper, we focus on robust impedance control against disturbances and parametric uncertainties within the nonlinear $\mathcal{H}_\infty$ optimal control framework—namely, nonlinear $\mathcal{H}_\infty$ impedance control. The nonlinear $\mathcal{H}_\infty$ optimal control problem is formulated as an $\mathcal{L}_2$-gain attenuation problem for a nonlinear closed-loop system and involves solving a partial differential equation or inequality, referred to as the Hamilton–Jacobi–Isaacs (HJI) equation or inequality [9], [10]. A key advantage of the $\mathcal{H}_\infty$ optimal control framework is that it explicitly incorporates unknown disturbances into the optimal control formulation. In other words, unlike other robust control techniques such as SMC, the $\mathcal{H}_\infty$ optimal control framework enables the controller to be designed by optimizing a performance index that accounts for unknown disturbances.

However, unlike the linear $\mathcal{H}_\infty$ optimal control problem, the nonlinear $\mathcal{H}_\infty$ optimal control problem remains a formidable challenge because the HJI equation is a nonlinear and multivariable partial differential equation. One common remedy is to apply linear $\mathcal{H}_\infty$ optimal control to nonlinear systems via linearization [11]. In the field of robotics, however, nonlinear control techniques such as the nonlinear DOB (NDOB) [12], [13] have often been preferred, as they explicitly account for the highly nonlinear nature of robot dynamics. In this context, various attempts have been made to solve the HJI equation directly for general nonlinear systems. Since no general solution has been found, most existing approaches rely on approximation techniques [14]–[16].

Fortunately, when we restrict our interest to robotic systems, a class of analytic solutions to the HJI equation can be found in [17]. The applications of the analytic solution, however, have been largely limited to trajectory tracking control [17], [18], and its extension to impedance control has rarely been explored [19], [20]. In the latter case, nonlinear $\mathcal{H}_\infty$ impedance control is achieved indirectly by solving the HJI equation for the deviation dynamics between the real robot and the simulated nominal robot. In contrast, building on the finding in [17], this paper derives the nonlinear $\mathcal{H}_\infty$ impedance control directly by solving the HJI equation for the interconnected robot–environment system.

A. Related work

In [17], a nonlinear $\mathcal{H}_\infty$ inverse-optimal trajectory tracking control for robotic systems (Euler–Lagrange systems) was proposed. The associated HJI equation was derived and

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Geonho Park is with the Department of Electrical and Electronic Engineering, Hanyang University, Ansan, 15588, South Korea. e-mail: ghpark@hanyang.ac.kr

Jonghoon Park is with Neuromeka, Seoul, 06023, South Korea. e-mail: coolcat@neuromeka.com

Youngjin Choi is with the Department of Robotics, Hanyang University, Ansan, 15588, South Korea. e-mail: cyj@hanyang.ac.kr

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solved analytically. The analytic solution to the nonlinear $\mathcal{H}_\infty$ inverse-optimal control problem leads to a control law referred to as *reference motion compensation* (RMC) with *reference error feedback* (REF). In [18], the robust stability of the REF-only control law, which takes a PID form, was formally established by extended disturbance input-to-state stability (ISS), and a performance analysis was also provided in terms of performance limitations and tuning. These studies addressed the trajectory tracking problem for robotic systems within the nonlinear $\mathcal{H}_\infty$ optimal control framework. Consequently, when physical interaction occurs, the interaction force is treated as a disturbance to be attenuated.

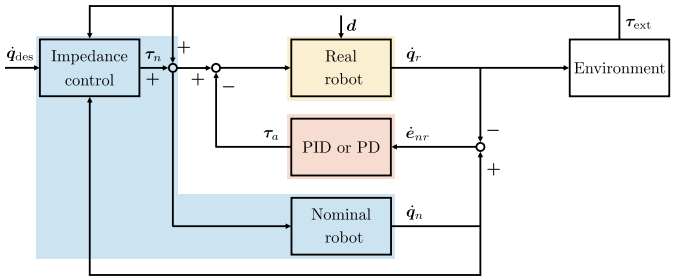
More recently, the nonlinear robust internal-loop compensator (NRIC) was proposed in [19]. The NRIC framework imparts nonlinear $\mathcal{H}_\infty$ optimality to any robot controller, provided that nominal stability is guaranteed. To this end, the NRIC framework adopts a two-loop control structure: In the outer loop, the nominal robot is simulated using the nominal controller through forward dynamics and numerical integration. In the inner loop, the real robot is forced to closely follow the simulated nominal robot via a high-gain joint space PID or PD controller. It was established that robust performance is guaranteed by nonlinear $\mathcal{H}_\infty$ optimality, while robust stability is ensured by extended disturbance ISS.

Given that the NRIC framework can impose nonlinear $\mathcal{H}_\infty$ optimality on any controller, impedance control implemented within the NRIC framework can be regarded as a nonlinear $\mathcal{H}_\infty$ impedance control. In [19], [20], a design of nonlinear $\mathcal{H}_\infty$ impedance control via the NRIC framework was proposed. In view of its inherent structure, however, it is essentially equivalent to admittance control (see Fig. 1). Hence, it reflects the fundamental nature of admittance causality, i.e., it is robust but not passive. Recent work [21] has developed a variant of the NRIC structure to address the non-passivity associated with the nonlinear $\mathcal{H}_\infty$ impedance control via the NRIC framework. Specifically, it introduces an additional feedback connection, referred to as bidirectional energy flow, into the NRIC structure to suppress internally generated energy during physical interaction. However, this approach achieves passivity by sacrificing its robustness and, in the extreme case, effectively reduces to pure impedance control, thereby becoming vulnerable to disturbances.

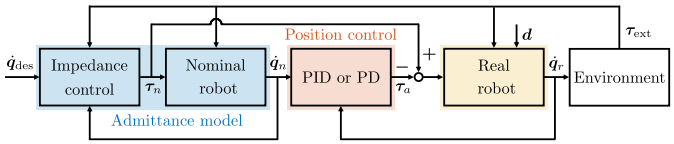
In fact, the robust internal-loop compensator (RIC) structure is essentially equivalent to the DOB [22], [23]. Therefore, its nonlinear extension, the NRIC, can be viewed as an NDOB [24], [25]. From this perspective, the nonlinear $\mathcal{H}_\infty$ impedance control via the NRIC framework can also be interpreted as an impedance control with the NDOB.

B. Contributions

To the best of the authors' knowledge, nonlinear $\mathcal{H}_\infty$ impedance control has rarely been reported in the literature. The existing nonlinear $\mathcal{H}_\infty$ impedance control via the NRIC framework exhibits admittance causality. Consequently, it can easily lead to oscillation and instability during physical interaction, particularly with stiff environments [1]–[3], [26]. In contrast, this paper formally establishes a nonlinear $\mathcal{H}_\infty$ impedance control that preserves impedance causality.



(a) Impedance control with NRIC framework



(b) Admittance control with feedforward input

Fig. 1. Equivalence of (a) impedance control with the NRIC framework and (b) admittance control with feedforward input.

First, the HJI equation for the interconnected robot–environment system is derived and solved analytically. The analytic solution leads to a control law that takes the form of RMC with REF, although the definition of the reference error extends that in [17]. Notably, the nonlinear $\mathcal{H}_\infty$ optimal control formulations in [17]–[19] require an artificial gain condition, as discussed later, since they are all based on the analytic solution proposed in [17]. In contrast, the analytic solution proposed in this paper eliminates this gain condition.

Second, robust stability of the interconnected robot–environment system is established by extended disturbance ISS under the proposed $\mathcal{H}_\infty$ impedance control. Consequently, the proposed $\mathcal{H}_\infty$ impedance control achieves robust stability and performance in rendering the desired dynamics in the presence of the extended disturbance.

Third, passivity of the interconnected robot–environment system is proven under the proposed $\mathcal{H}_\infty$ impedance control. Consequently, the proposed $\mathcal{H}_\infty$ impedance control realizes robust and passive physical interaction.

The remainder of this paper is organized as follows: Section II reviews the nonlinear $\mathcal{H}_\infty$ inverse-optimal control via RMC with REF in [17]. Section III presents the proposed nonlinear $\mathcal{H}_\infty$ impedance control. Section IV establishes the robust stability of the proposed $\mathcal{H}_\infty$ impedance control. Section V provides a passivity analysis of the proposed $\mathcal{H}_\infty$ impedance control. Section VI extends the proposed $\mathcal{H}_\infty$ impedance control framework to the operational space formulation. In Section VII, experimental results are presented, followed by the concluding remarks in Section VIII. Throughout this paper, $\widehat{(\cdot)}$ represents the estimated or measured value of $(\cdot)$, and boldface type is used to represent vectors and matrices, whereas scalar quantities are written in normal font. The notations $\mathbf{0}$ and $\boldsymbol{I}$ denote the zero vector or matrix and the identity matrix of appropriate dimensions, respectively.

II. Nonlinear $\mathcal{H}_\infty$ Inverse-Optimal Control

This section briefly reviews [17], namely, the nonlinear $\mathcal{H}_\infty$ inverse-optimal position control via RMC with REF.

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A. Nonlinear $\mathcal{H}_\infty$ optimal control problem

Consider a nonlinear control-affine and disturbance-affine system with control input $\boldsymbol{u}$ and disturbance input $\boldsymbol{w}$:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x},t) + \boldsymbol{g}(\boldsymbol{x},t)\boldsymbol{u} + \boldsymbol{p}(\boldsymbol{x},t)\boldsymbol{w}, \tag{1}$$

with $\boldsymbol{f}(\boldsymbol{x}^*,t)=\boldsymbol{0}$ at the equilibrium $\boldsymbol{x}^*=\boldsymbol{0}$. For the system (1), consider the performance index

$$J(\boldsymbol{x}(t),t,\boldsymbol{u}(\sigma)_{t\le\sigma\le T},\boldsymbol{w}(\sigma)_{t\le\sigma\le T}) = m(\boldsymbol{x}(T),T) + \int_t^T l(\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},\sigma)\,d\sigma,$$

where the running cost and the terminal cost are given as

$$l(\boldsymbol{x},\boldsymbol{u},\boldsymbol{w},t) = \frac{1}{2}\left\{\boldsymbol{z}^T(\boldsymbol{x},\boldsymbol{u},t)\boldsymbol{z}(\boldsymbol{x},\boldsymbol{u},t) - \gamma^2\boldsymbol{w}^T\boldsymbol{w}\right\},$$
$$m(\boldsymbol{x}(T),T) = V(\boldsymbol{x}(T),T),$$

with the cost variable $\boldsymbol{z}$ defined by

$$\boldsymbol{z}(\boldsymbol{x},\boldsymbol{u},t) = \boldsymbol{h}(\boldsymbol{x},t) + \boldsymbol{k}(\boldsymbol{x},t)\boldsymbol{u}. \tag{2}$$

Here, the disturbance $\boldsymbol{w}$ reduces the running cost in proportion to γ^2. It is assumed that (a) $\boldsymbol{f}(\boldsymbol{x},t)$, $\boldsymbol{g}(\boldsymbol{x},t)$, and $\boldsymbol{p}(\boldsymbol{x},t)$ are sufficiently smooth, (b) $\boldsymbol{g}(\boldsymbol{x},t)$ and $\boldsymbol{p}(\boldsymbol{x},t)$ have full column rank, (c) $\boldsymbol{h}^T(\boldsymbol{x},t)\boldsymbol{k}(\boldsymbol{x},t)=\boldsymbol{0}$ and $\boldsymbol{k}^T(\boldsymbol{x},t)\boldsymbol{k}(\boldsymbol{x},t)>\boldsymbol{0}$, and (d) $\boldsymbol{h}(\boldsymbol{x}^*,t)=\boldsymbol{0}$. The nonlinear $\mathcal{H}_\infty$ optimal control problem is stated as follows. Given a desired $\mathcal{L}_2$-gain $\gamma>0$, find a state feedback control $\boldsymbol{u}=\boldsymbol{\alpha}(\boldsymbol{x},t)$ with $\boldsymbol{\alpha}(\boldsymbol{x}^*,t)=\boldsymbol{0}$ such that the following $\mathcal{L}_2$-gain attenuation requirement is satisfied for the closed-loop system:

$$\int_0^T \boldsymbol{z}^T(\boldsymbol{x},\sigma)\boldsymbol{z}(\boldsymbol{x},\sigma)\,d\sigma \le \gamma^2\int_0^T \boldsymbol{w}^T\boldsymbol{w}\,d\sigma \tag{3}$$

for all $T\ge 0$ and for all $\boldsymbol{w}\in\mathcal{L}_2[0,T]$, with $\boldsymbol{x}(0)=\boldsymbol{0}$.[1]

It is well known that the $\mathcal{L}_2$-gain attenuation problem (3) reduces to solving a partial differential equation, called the Hamilton–Jacobi–Isaacs (HJI) equation, which takes the form

$$V_t(\boldsymbol{x},t) + \boldsymbol{V}_x(\boldsymbol{x},t)\boldsymbol{f}(\boldsymbol{x},t) + \frac{1}{2}\boldsymbol{h}^T(\boldsymbol{x},t)\boldsymbol{h}(\boldsymbol{x},t) - \frac{1}{2}\boldsymbol{V}_x(\boldsymbol{x},t)\Big[\boldsymbol{g}(\boldsymbol{x},t)\left\{\boldsymbol{k}^T(\boldsymbol{x},t)\boldsymbol{k}(\boldsymbol{x},t)\right\}^{-1}\boldsymbol{g}^T(\boldsymbol{x},t) - \frac{1}{\gamma^2}\boldsymbol{p}(\boldsymbol{x},t)\boldsymbol{p}^T(\boldsymbol{x},t)\Big]\boldsymbol{V}_x^T(\boldsymbol{x},t) = 0, \tag{4}$$

with the terminal condition $V(\boldsymbol{x}(T),T)=m(\boldsymbol{x}(T),T)$,[2] where $V_t(\boldsymbol{x},t)=\partial V(\boldsymbol{x},t)/\partial t$ and $\boldsymbol{V}_x(\boldsymbol{x},t)=\partial V(\boldsymbol{x},t)/\partial \boldsymbol{x}^T$. If there exists a continuously differentiable function $V(\boldsymbol{x},t)\ge 0$ with $V(\boldsymbol{x}^*,t)=0$ satisfying the HJI equation (4), the optimal control $\boldsymbol{u}$ that satisfies the $\mathcal{L}_2$-gain attenuation (3) is given by

$$\boldsymbol{u} = -\left\{\boldsymbol{k}^T(\boldsymbol{x},t)\boldsymbol{k}(\boldsymbol{x},t)\right\}^{-1}\boldsymbol{g}^T(\boldsymbol{x},t)\boldsymbol{V}_x^T(\boldsymbol{x},t).$$

However, solving the HJI equation (4) is generally a very challenging problem, as it is a nonlinear and multivariable partial differential equation. An alternative to overcome this difficulty is to take an inverse approach [17]. In standard optimal control problems, the solution function $V(\boldsymbol{x},t)$ is to be found for the prescribed cost functions $\boldsymbol{h}(\boldsymbol{x},t)$ and $\boldsymbol{k}(\boldsymbol{x},t)$. In contrast, in inverse-optimal control problems, we seek the solution function $V(\boldsymbol{x},t)$ as well as the optimal cost functions $\boldsymbol{h}(\boldsymbol{x},t)$ and $\boldsymbol{k}(\boldsymbol{x},t)$, such that the HJI equation is satisfied.

B. Nonlinear $\mathcal{H}_\infty$ inverse-optimal control via RMC with REF

The equations of motion of an Euler–Lagrange system with n degrees of freedom (DOF) are given by

$$\boldsymbol{\tau} + \boldsymbol{\tau}_{\text{ext}} = \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{g}(\boldsymbol{q}) + \boldsymbol{d}(\boldsymbol{q},\dot{\boldsymbol{q}},t), \tag{5}$$

where $\boldsymbol{q}\in\mathbb{R}^n$ denotes the generalized coordinates of the system, $\boldsymbol{\tau}\in\mathbb{R}^n$ denotes the control inputs, and $\boldsymbol{\tau}_{\text{ext}}\in\mathbb{R}^n$ represents the external forces; $\boldsymbol{M}(\boldsymbol{q})\in\mathbb{R}^{n\times n}$ denotes the inertia matrix, $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\in\mathbb{R}^{n\times n}$ denotes the Coriolis/centrifugal matrix, $\boldsymbol{g}(\boldsymbol{q})\in\mathbb{R}^n$ denotes the gravitational forces, and $\boldsymbol{d}(\boldsymbol{q},\dot{\boldsymbol{q}},t)\in\mathbb{R}^n$ stands for the disturbance arising from unmodeled dynamics, including friction. The Euler–Lagrange system has the property

$$\dot{\boldsymbol{M}}(\boldsymbol{q},\dot{\boldsymbol{q}}) = \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}}) + \boldsymbol{C}^T(\boldsymbol{q},\dot{\boldsymbol{q}}), \tag{6}$$

which is equivalent to $\dot{\boldsymbol{M}}-2\boldsymbol{C}$ being skew-symmetric.

The control design begins with the following form:

$$\boldsymbol{\tau} = \widehat{\boldsymbol{M}}(\boldsymbol{q})\ddot{\boldsymbol{q}}_{\text{ref}} + \widehat{\boldsymbol{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}_{\text{ref}} + \widehat{\boldsymbol{g}}(\boldsymbol{q}) - \boldsymbol{u}, \tag{7}$$

where the reference acceleration and velocity for trajectory tracking are designed as

$$\ddot{\boldsymbol{q}}_{\text{ref}} = \ddot{\boldsymbol{q}}_{\text{des}} + k_d\dot{\boldsymbol{e}} + k_p\boldsymbol{e},$$
$$\dot{\boldsymbol{q}}_{\text{ref}} = \dot{\boldsymbol{q}}_{\text{des}} + k_d\boldsymbol{e} + k_p\int \boldsymbol{e},$$

in which $\boldsymbol{e}=\boldsymbol{q}_{\text{des}}-\boldsymbol{q}$ denotes the configuration error, and $k_d>0$ and $k_p>0$ are the constant derivative and proportional gains, respectively. This modified computed torque control (7) is called reference motion compensation (RMC) in [17].

Substituting the control (7) into the system dynamics (5) and assuming[3] $\boldsymbol{\tau}_{\text{ext}}=\boldsymbol{0}$ yields

$$\boldsymbol{u} + \boldsymbol{w} = \boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{e}}_{\text{ref}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{e}}_{\text{ref}}, \tag{8}$$

where $\ddot{\boldsymbol{e}}_{\text{ref}}=\ddot{\boldsymbol{q}}_{\text{ref}}-\ddot{\boldsymbol{q}}$, $\dot{\boldsymbol{e}}_{\text{ref}}=\dot{\boldsymbol{q}}_{\text{ref}}-\dot{\boldsymbol{q}}$, and the extended disturbance $\boldsymbol{w}$ is defined by

$$\boldsymbol{w} = \widetilde{\boldsymbol{M}}(\boldsymbol{q})\ddot{\boldsymbol{q}}_{\text{ref}} + \widetilde{\boldsymbol{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}_{\text{ref}} + \widetilde{\boldsymbol{g}}(\boldsymbol{q}) + \boldsymbol{d}(\boldsymbol{q},\dot{\boldsymbol{q}},t),$$

with parametric uncertainties $\widetilde{\boldsymbol{M}}=\boldsymbol{M}-\widehat{\boldsymbol{M}}$, $\widetilde{\boldsymbol{C}}=\boldsymbol{C}-\widehat{\boldsymbol{C}}$, and $\widetilde{\boldsymbol{g}}=\boldsymbol{g}-\widehat{\boldsymbol{g}}$. The system (8) can be expressed in state space with state $\boldsymbol{x}=\begin{bmatrix}\boldsymbol{x}_1^T & \boldsymbol{x}_2^T & \boldsymbol{x}_3^T\end{bmatrix}^T=\begin{bmatrix}\int \boldsymbol{e}^T & \boldsymbol{e}^T & \dot{\boldsymbol{e}}^T\end{bmatrix}^T\in\mathbb{R}^{3n}$ as

$$\dot{\boldsymbol{x}} = \boldsymbol{A}(\boldsymbol{x},t)\boldsymbol{x} + \boldsymbol{B}(\boldsymbol{x},t)\boldsymbol{u} + \boldsymbol{B}(\boldsymbol{x},t)\boldsymbol{w}, \tag{9}$$

where

$$\boldsymbol{A} = \begin{bmatrix}\boldsymbol{0} & \boldsymbol{I} & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I}\\ -k_p\boldsymbol{M}^{-1}\boldsymbol{C} & -k_d\boldsymbol{M}^{-1}\boldsymbol{C}-k_p\boldsymbol{I} & -\boldsymbol{M}^{-1}\boldsymbol{C}-k_d\boldsymbol{I}\end{bmatrix},$$
$$\boldsymbol{B} = \begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{0}\\ \boldsymbol{M}^{-1}\end{bmatrix}.$$

[1] The nonzero initial condition case follows directly from the zero initial condition case.

[2] The terminal cost is chosen as the solution function itself, so the terminal condition is satisfied by construction.

[3] In [17], $\boldsymbol{\tau}_{\text{ext}}$ is not explicitly considered in the formulation. Therefore, when $\boldsymbol{\tau}_{\text{ext}}$ is present, it is lumped into the extended disturbance $\boldsymbol{w}$ and treated as a disturbance to be attenuated.

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Assuming that the cost functions $\boldsymbol{h}(\boldsymbol{x},t)$ and $\boldsymbol{k}(\boldsymbol{x},t)$ in (2) satisfy $\boldsymbol{h}^T(\boldsymbol{x},t)\boldsymbol{h}(\boldsymbol{x},t)=\boldsymbol{x}^T\boldsymbol{Q}(\boldsymbol{x},t)\boldsymbol{x}$ and $\boldsymbol{k}^T(\boldsymbol{x},t)\boldsymbol{k}(\boldsymbol{x},t)=\boldsymbol{R}(\boldsymbol{x},t)$, and that the solution to the HJI equation (4) for the system (9) is given by $V(\boldsymbol{x},t)=(1/2)\boldsymbol{x}^T\boldsymbol{P}(\boldsymbol{x},t)\boldsymbol{x}$, where $\boldsymbol{P}(\boldsymbol{x},t)$ is not an explicit function of $\boldsymbol{x}_3=\dot{\boldsymbol{e}}$, the HJI equation reduces to a matrix differential Riccati equation (DRE) as follows [17]:

$$\dot{\boldsymbol{P}}(\boldsymbol{x},t)+\boldsymbol{A}^T(\boldsymbol{x},t)\boldsymbol{P}(\boldsymbol{x},t)+\boldsymbol{P}(\boldsymbol{x},t)\boldsymbol{A}(\boldsymbol{x},t)+\boldsymbol{Q}(\boldsymbol{x},t) - \boldsymbol{P}(\boldsymbol{x},t)\boldsymbol{B}(\boldsymbol{x},t)\underbrace{\left\{\boldsymbol{R}^{-1}(\boldsymbol{x},t)-\frac{1}{\gamma^2}\boldsymbol{I}\right\}}_{\boldsymbol{K}_\gamma}\boldsymbol{B}^T(\boldsymbol{x},t)\boldsymbol{P}(\boldsymbol{x},t)=\boldsymbol{0}. \tag{10}$$

If there exists a solution matrix $\boldsymbol{P}(\boldsymbol{x},t)\geq\boldsymbol{0}$ to the DRE (10), then the nonlinear $\mathcal{H}_\infty$ optimal control satisfying the $\mathcal{L}_2$-gain attenuation requirement

$$\int_0^T\left\{\boldsymbol{x}^T\boldsymbol{Q}(\boldsymbol{x},\sigma)\boldsymbol{x}+\boldsymbol{u}^T\boldsymbol{R}(\boldsymbol{x},\sigma)\boldsymbol{u}\right\}d\sigma \leq\gamma^2\int_0^T\left\{\boldsymbol{w}^T(\boldsymbol{x},\sigma)\boldsymbol{w}(\boldsymbol{x},\sigma)\right\}d\sigma \tag{11}$$

is given by

$$\boldsymbol{u}=-\boldsymbol{R}^{-1}(\boldsymbol{x},t)\boldsymbol{B}^T(\boldsymbol{x},t)\boldsymbol{P}(\boldsymbol{x},t)\boldsymbol{x}.$$

In [17], an analytic solution to the DRE (10) with $\boldsymbol{P}(\boldsymbol{x},t)>\boldsymbol{0}$, $\boldsymbol{Q}>\boldsymbol{0}$, and $\boldsymbol{R}>\boldsymbol{0}$ was obtained via an inverse approach by exploiting the property (6) as follows (for the proof, see [17]):

$$\begin{aligned}\boldsymbol{P}(\boldsymbol{x},t)&=\alpha\begin{bmatrix}k_p^2\boldsymbol{M}+\alpha k_pk_d\boldsymbol{K}_\gamma & k_pk_d\boldsymbol{M}+\alpha k_p\boldsymbol{K}_\gamma & k_p\boldsymbol{M}\\ k_pk_d\boldsymbol{M}+\alpha k_p\boldsymbol{K}_\gamma & k_d^2\boldsymbol{M}+\alpha k_d\boldsymbol{K}_\gamma & k_d\boldsymbol{M}\\ k_p\boldsymbol{M} & k_d\boldsymbol{M} & \boldsymbol{M}\end{bmatrix},\\ \boldsymbol{Q}&=\alpha^2\begin{bmatrix}k_p^2\boldsymbol{K}_\gamma & \boldsymbol{0} & \boldsymbol{0}\\ \boldsymbol{0} & (k_d^2-2k_p)\boldsymbol{K}_\gamma & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{K}_\gamma\end{bmatrix},\\ \boldsymbol{R}&=\left(\boldsymbol{K}_\gamma+\frac{1}{\gamma^2}\boldsymbol{I}\right)^{-1}\end{aligned} \tag{12}$$

under the gain condition

$$k_d^2>2k_p, \tag{13}$$

where $\alpha>0$ is a positive constant scalar and $\boldsymbol{K}_\gamma>\boldsymbol{0}$ is a positive definite constant matrix. The resulting $\mathcal{H}_\infty$ optimal control is given by

$$\boldsymbol{u}=-\alpha\left(\boldsymbol{K}_\gamma+\frac{1}{\gamma^2}\boldsymbol{I}\right)\dot{\boldsymbol{e}}_{\rm ref}, \tag{14}$$

where

$$\dot{\boldsymbol{e}}_{\rm ref}=\dot{\boldsymbol{e}}+k_d\boldsymbol{e}+k_p\int\boldsymbol{e}$$

is referred to as the reference error, and thus the resulting optimal control (14) is called reference error feedback (REF) in [17]. Substituting (14) into (7) yields the final form of the nonlinear $\mathcal{H}_\infty$ inverse-optimal position control as

$$\boldsymbol{\tau}=\widehat{\boldsymbol{M}}(\boldsymbol{q})\ddot{\boldsymbol{q}}_{\rm ref}+\widehat{\boldsymbol{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}_{\rm ref}+\widehat{\boldsymbol{g}}(\boldsymbol{q})+\alpha\left(\boldsymbol{K}_\gamma+\frac{1}{\gamma^2}\boldsymbol{I}\right)\dot{\boldsymbol{e}}_{\rm ref}. \tag{15}$$

This control form is called RMC with REF in [17].

Notably, although the analytic solution $\boldsymbol{P}(\boldsymbol{x},t)$ in (12) explicitly depends on the true inertia matrix, which is unknown in practice, this dependence is completely canceled out in the optimal control $\boldsymbol{u}=-\boldsymbol{R}^{-1}\boldsymbol{B}^T(\boldsymbol{x},t)\boldsymbol{P}(\boldsymbol{x},t)\boldsymbol{x}$. The resulting control law (15) relies solely on the nominal dynamics. In other words, the underlying philosophy of [17] is that the $\mathcal{H}_\infty$ optimal control can be constructed without exact knowledge of the true model, using only the nominal model.

III. Nonlinear $\mathcal{H}_\infty$ Impedance Control

Impedance control aims to render the closed-loop dynamics as the following desired impedance model:

$$\boldsymbol{\Lambda}\ddot{\boldsymbol{e}}+\boldsymbol{D}\dot{\boldsymbol{e}}+\boldsymbol{K}\boldsymbol{e}=-\boldsymbol{\tau}_{\rm ext}, \tag{16}$$

where $\boldsymbol{\Lambda}>\boldsymbol{0}$, $\boldsymbol{D}\geq\boldsymbol{0}$, and $\boldsymbol{K}\geq\boldsymbol{0}$ are constant diagonal matrices specifying the desired inertia, damping, and stiffness, respectively. To this end, the control law can be designed via feedback linearization as

$$\boldsymbol{\tau}=\widehat{\boldsymbol{M}}(\boldsymbol{q})\ddot{\boldsymbol{q}}_{\rm ref}+\widehat{\boldsymbol{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}+\widehat{\boldsymbol{g}}(\boldsymbol{q})-\widehat{\boldsymbol{\tau}}_{\rm ext}, \tag{17}$$

where the reference acceleration is given by

$$\ddot{\boldsymbol{q}}_{\rm ref}=\ddot{\boldsymbol{q}}_{\rm des}+\boldsymbol{\Lambda}^{-1}\left(\boldsymbol{D}\dot{\boldsymbol{e}}+\boldsymbol{K}\boldsymbol{e}+\widehat{\boldsymbol{\tau}}_{\rm ext}\right).$$

Assuming a perfect dynamic model, an ideal force sensor, and no disturbances, i.e., $\boldsymbol{M}=\widehat{\boldsymbol{M}}$, $\boldsymbol{C}=\widehat{\boldsymbol{C}}$, $\boldsymbol{g}=\widehat{\boldsymbol{g}}$, $\boldsymbol{\tau}_{\rm ext}=\widehat{\boldsymbol{\tau}}_{\rm ext}$, and $\boldsymbol{d}=\boldsymbol{0}$, the closed-loop dynamics reduces to (16). However, in practical situations, joint friction and inevitable parametric uncertainties are present, and the force sensor is inherently noisy, so that the rendered behavior may deviate significantly from the desired behavior. Robustness in impedance control means that the closed-loop dynamics remains close to the desired impedance model (16) despite these disturbances and uncertainties. This paper aims to achieve this objective within the nonlinear $\mathcal{H}_\infty$ optimal control framework.

The control design begins with the RMC:

$$\boldsymbol{\tau}=\widehat{\boldsymbol{M}}(\boldsymbol{q})\ddot{\boldsymbol{q}}_{\rm ref}+\widehat{\boldsymbol{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}_{\rm ref}+\widehat{\boldsymbol{g}}(\boldsymbol{q})-\widehat{\boldsymbol{\tau}}_{\rm ext}-\boldsymbol{u}, \tag{18}$$

where the reference acceleration and velocity are designed as

$$\begin{aligned}\ddot{\boldsymbol{q}}_{\rm ref}&=\ddot{\boldsymbol{q}}_{\rm des}+\boldsymbol{\Lambda}^{-1}\left(\boldsymbol{D}\dot{\boldsymbol{e}}+\boldsymbol{K}\boldsymbol{e}+\widehat{\boldsymbol{\tau}}_{\rm ext}\right),\\ \dot{\boldsymbol{q}}_{\rm ref}&=\dot{\boldsymbol{q}}_{\rm des}+\boldsymbol{\Lambda}^{-1}\left(\boldsymbol{D}\boldsymbol{e}+\boldsymbol{K}\int\boldsymbol{e}+\int\widehat{\boldsymbol{\tau}}_{\rm ext}\right).\end{aligned}$$

Substituting the control (18) into the system dynamics (5) yields

$$\boldsymbol{u}+\boldsymbol{w}=\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{e}}_{\rm ref}+\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{e}}_{\rm ref}, \tag{19}$$

where $\ddot{\boldsymbol{e}}_{\rm ref}=\ddot{\boldsymbol{q}}_{\rm ref}-\ddot{\boldsymbol{q}}$, $\dot{\boldsymbol{e}}_{\rm ref}=\dot{\boldsymbol{q}}_{\rm ref}-\dot{\boldsymbol{q}}$, and the extended disturbance $\boldsymbol{w}$ is defined by

$$\boldsymbol{w}=\widetilde{\boldsymbol{M}}(\boldsymbol{q})\ddot{\boldsymbol{q}}_{\rm ref}+\widetilde{\boldsymbol{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}_{\rm ref}+\widetilde{\boldsymbol{g}}(\boldsymbol{q})+\boldsymbol{d}(\boldsymbol{q},\dot{\boldsymbol{q}},t)-\widetilde{\boldsymbol{\tau}}_{\rm ext}, \tag{20}$$

with the force sensor noise $\widetilde{\boldsymbol{\tau}}_{\rm ext}=\boldsymbol{\tau}_{\rm ext}-\widehat{\boldsymbol{\tau}}_{\rm ext}$.

Since the $\mathcal{L}_2$-gain attenuation problem (3) is formulated with respect to the extended disturbance $\boldsymbol{w}$ (20), a proper characterization of $\boldsymbol{w}$ is necessary. Importantly, the extended disturbance must be bounded; otherwise, the optimality criterion (3) becomes vacuous. Therefore, the following mild assumptions are introduced to account for unknown disturbances and environments.

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Assumption 1: The disturbance $\boldsymbol{d}$ is bounded and satisfies

$$-\dot{\boldsymbol{q}}^T \boldsymbol{d} \leq 0,$$

which implies that the disturbance arises from passive sources such as friction, i.e., it does not inject energy into the system.

This assumption is considered to be the minimal information required about unknown disturbances.

Assumption 2: The environment is modeled as a mechanical impedance

$$\boldsymbol{\Lambda}_{\mathrm{env}} \ddot{\boldsymbol{\delta}} + \boldsymbol{D}_{\mathrm{env}} \dot{\boldsymbol{\delta}} + \boldsymbol{K}_{\mathrm{env}} \boldsymbol{\delta} = \boldsymbol{\tau}_{\mathrm{ext}}, \tag{21}$$

where $\boldsymbol{\delta} = \boldsymbol{q}_{\mathrm{env}} - \boldsymbol{q}$ denotes the environmental deformation or displacement, and $\boldsymbol{q}_{\mathrm{env}}$ denotes the (possibly time-varying) neutral configuration of the environment; $\boldsymbol{\Lambda}_{\mathrm{env}} > \boldsymbol{0}$, $\boldsymbol{D}_{\mathrm{env}} \geq \boldsymbol{0}$, and $\boldsymbol{K}_{\mathrm{env}} \geq \boldsymbol{0}$ are unknown but bounded constant matrices. In the absence of interaction, the environment moves in a bounded manner, i.e., $\boldsymbol{q}_{\mathrm{env}}$, $\dot{\boldsymbol{q}}_{\mathrm{env}}$, and $\ddot{\boldsymbol{q}}_{\mathrm{env}}$ remain bounded.

This is a common assumption because stabilizing an interconnected system with an active component poses significant challenges. Since the environmental parameters are unknown and the actual external force $\boldsymbol{\tau}_{\mathrm{ext}}$ is generally unavailable, an additional assumption is introduced.

Assumption 3: The force signal measured by the force sensor is modeled by the following dynamics:

$$\begin{aligned}\boldsymbol{\tau}_{\mathrm{ext}} &= \boldsymbol{\Lambda}_{\mathrm{sen}} \ddot{\boldsymbol{\epsilon}} + \boldsymbol{D}_{\mathrm{sen}} \dot{\boldsymbol{\epsilon}} + \boldsymbol{K}_{\mathrm{sen}} \boldsymbol{\epsilon} \\ &= \underbrace{(\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}} \ddot{\boldsymbol{\epsilon}} + \widehat{\boldsymbol{D}}_{\mathrm{sen}} \dot{\boldsymbol{\epsilon}} + \widehat{\boldsymbol{K}}_{\mathrm{sen}} \boldsymbol{\epsilon})}_{\widehat{\boldsymbol{\tau}}_{\mathrm{ext}}} + \underbrace{(\widetilde{\boldsymbol{\Lambda}}_{\mathrm{sen}} \ddot{\boldsymbol{\epsilon}} + \widetilde{\boldsymbol{D}}_{\mathrm{sen}} \dot{\boldsymbol{\epsilon}} + \widetilde{\boldsymbol{K}}_{\mathrm{sen}} \boldsymbol{\epsilon})}_{\widetilde{\boldsymbol{\tau}}_{\mathrm{ext}}},\end{aligned} \tag{22}$$

where $\widehat{\boldsymbol{\tau}}_{\mathrm{ext}}$ denotes the measured force signal, $\widetilde{\boldsymbol{\tau}}_{\mathrm{ext}}$ denotes the force sensor noise, and $\boldsymbol{\epsilon}$ denotes the sensor-internal displacement; $\boldsymbol{\Lambda}_{\mathrm{sen}} > \boldsymbol{0}$, $\boldsymbol{D}_{\mathrm{sen}} \geq \boldsymbol{0}$, and $\boldsymbol{K}_{\mathrm{sen}} \geq \boldsymbol{0}$ are unknown but bounded (possibly time-varying) matrices. The measured force signal $\widehat{\boldsymbol{\tau}}_{\mathrm{ext}}$ is bounded.

The sensor dynamics (22) can be viewed as virtual dynamics rather than the true physical sensor dynamics, provided that $\boldsymbol{\Lambda}_{\mathrm{env}} \ddot{\boldsymbol{\delta}} + \boldsymbol{D}_{\mathrm{env}} \dot{\boldsymbol{\delta}} + \boldsymbol{K}_{\mathrm{env}} \boldsymbol{\delta} = \boldsymbol{\tau}_{\mathrm{ext}} = \boldsymbol{\Lambda}_{\mathrm{sen}} \ddot{\boldsymbol{\epsilon}} + \boldsymbol{D}_{\mathrm{sen}} \dot{\boldsymbol{\epsilon}} + \boldsymbol{K}_{\mathrm{sen}} \boldsymbol{\epsilon}$ holds. For any given $\boldsymbol{\delta}$, $\dot{\boldsymbol{\delta}}$, and $\ddot{\boldsymbol{\delta}}$, a displacement $\boldsymbol{\epsilon}$ and its derivatives satisfying the above dynamics always exist (although unknown). Thus, this assumption is considered reasonable.

Since the sensor dynamics (22) approximates the environmental dynamics (21), it can be regarded as the nominal environmental dynamics. The nominal environmental parameters $\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}} > \boldsymbol{0}$, $\widehat{\boldsymbol{D}}_{\mathrm{sen}} \geq \boldsymbol{0}$, and $\widehat{\boldsymbol{K}}_{\mathrm{sen}} \geq \boldsymbol{0}$ can be freely chosen (preferably chosen to approximate the environmental parameters), since the modeling error $\widetilde{\boldsymbol{\Lambda}}_{\mathrm{sen}} \ddot{\boldsymbol{\epsilon}} + \widetilde{\boldsymbol{D}}_{\mathrm{sen}} \dot{\boldsymbol{\epsilon}} + \widetilde{\boldsymbol{K}}_{\mathrm{sen}} \boldsymbol{\epsilon} = \widetilde{\boldsymbol{\tau}}_{\mathrm{ext}}$ is explicitly included in the extended disturbance $\boldsymbol{w}$ (20). Once these parameters are chosen, a displacement $\boldsymbol{\epsilon}$ and its derivatives satisfying the dynamics (22) always exist for any piecewise continuous measured force signal $\widehat{\boldsymbol{\tau}}_{\mathrm{ext}}$.

Let us now investigate the boundedness of the extended disturbance $\boldsymbol{w}$. First, $||\widetilde{\boldsymbol{M}}(\boldsymbol{q})|| \leq m_0$ and $||\widetilde{\boldsymbol{g}}(\boldsymbol{q})|| \leq g_0$ hold for some constants $m_0 \geq 0$ and $g_0 \geq 0$. Second, $||\widetilde{\boldsymbol{C}}(\boldsymbol{q}, \dot{\boldsymbol{q}})|| \leq c_0 ||\dot{\boldsymbol{q}}||$ holds for some constant $c_0 \geq 0$. It should be noted that in the second term on the right-hand side of (20), $\dot{\boldsymbol{q}}_{\mathrm{ref}}$ originating from the RMC appears instead of $\dot{\boldsymbol{q}}$, which prevents finite-time divergence due to the quadratic velocity term. If the RMC (18) is replaced with the conventional computed torque control form as in (17), this risk remains. Third, the desired trajectory $\boldsymbol{q}_{\mathrm{des}}$, $\dot{\boldsymbol{q}}_{\mathrm{des}}$, and $\ddot{\boldsymbol{q}}_{\mathrm{des}}$ can be specified to be bounded. Fourth, by Assumption 1, $\boldsymbol{d}$ is bounded. Fifth, by Assumptions 2 and 3, $\widetilde{\boldsymbol{\tau}}_{\mathrm{ext}} = \boldsymbol{\tau}_{\mathrm{ext}} - \widehat{\boldsymbol{\tau}}_{\mathrm{ext}}$ is bounded if $\boldsymbol{\delta}$, $\dot{\boldsymbol{\delta}}$, and $\ddot{\boldsymbol{\delta}}$ are bounded. Hence, the following can be stated.

Remark 1: Given Assumptions 1, 2, and 3, the extended disturbance $\boldsymbol{w}$ remains bounded, provided that $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\dot{\boldsymbol{q}}_{\mathrm{ref}}$, $\boldsymbol{\delta}$, $\dot{\boldsymbol{\delta}}$, and $\ddot{\boldsymbol{\delta}}$ remain bounded. The boundedness of $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\boldsymbol{\delta}$, and $\dot{\boldsymbol{\delta}}$ will be established for the closed-loop system in Corollary 1 and Remark 6; that of $\dot{\boldsymbol{q}}_{\mathrm{ref}}$ then follows from $\dot{\boldsymbol{q}}_{\mathrm{ref}} = \dot{\boldsymbol{e}}_{\mathrm{ref}} + \dot{\boldsymbol{q}}$, with the boundedness of $\dot{\boldsymbol{e}}_{\mathrm{ref}}$ established in Theorem 2; and that of $\ddot{\boldsymbol{\delta}}$ then follows from the dynamics (5) and (21). Accordingly, the extended disturbance $\boldsymbol{w}$ is treated as bounded throughout this paper.

Based on the preceding discussion, the proposed control law is presented in the following theorem.

Theorem 1: For an Euler–Lagrange system (5) satisfying Assumption 1 that physically interacts with an environment (21) satisfying Assumptions 2 and 3, if the nominal environmental parameters are chosen as constant matrices satisfying the impedance matching condition with the desired impedance in an inertia-normalized sense, i.e.,

$$\widehat{\boldsymbol{D}}_{\mathrm{sen}} = \boldsymbol{D} \boldsymbol{\Lambda}^{-1} \widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}, \quad \widehat{\boldsymbol{K}}_{\mathrm{sen}} = \boldsymbol{K} \boldsymbol{\Lambda}^{-1} \widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}, \tag{23}$$

then the nonlinear $\mathcal{H}_\infty$ optimal control satisfying the $\mathcal{L}_2$-gain attenuation requirement (11) for the interconnected system of the Euler–Lagrange system and the environment is given by

$$\boldsymbol{u} = -\alpha \left(\boldsymbol{K}_\gamma + \frac{1}{\gamma^2} \boldsymbol{I} \right) \dot{\boldsymbol{e}}_{\mathrm{ref}},$$

with a constant scalar $\alpha > 0$, a constant diagonal gain matrix $\boldsymbol{K}_\gamma > \boldsymbol{0}$, the desired $\mathcal{L}_2$-gain $\gamma > 0$, and the reference error defined by

$$\dot{\boldsymbol{e}}_{\mathrm{ref}} = \dot{\boldsymbol{e}} + \boldsymbol{\Lambda}^{-1} \left(\boldsymbol{D} \boldsymbol{e} + \boldsymbol{K} \int \boldsymbol{e} + \int \widehat{\boldsymbol{\tau}}_{\mathrm{ext}} \right).$$

The control law takes the form of the REF.

Proof: The Euler–Lagrange system (19) can be rearranged to give

$$\begin{aligned}\ddot{\boldsymbol{e}} = &- \left(\boldsymbol{\Lambda}^{-1} \boldsymbol{D} + \boldsymbol{M}^{-1} \boldsymbol{C} \right) \dot{\boldsymbol{e}} - \left(\boldsymbol{\Lambda}^{-1} \boldsymbol{K} + \boldsymbol{M}^{-1} \boldsymbol{C} \boldsymbol{\Lambda}^{-1} \boldsymbol{D} \right) \boldsymbol{e} \\ &- \boldsymbol{M}^{-1} \boldsymbol{C} \boldsymbol{\Lambda}^{-1} \boldsymbol{K} \int \boldsymbol{e} - \boldsymbol{\Lambda}^{-1} \widehat{\boldsymbol{\tau}}_{\mathrm{ext}} - \boldsymbol{M}^{-1} \boldsymbol{C} \boldsymbol{\Lambda}^{-1} \int \widehat{\boldsymbol{\tau}}_{\mathrm{ext}} \\ &+ \boldsymbol{M}^{-1} \boldsymbol{u} + \boldsymbol{M}^{-1} \boldsymbol{w}.\end{aligned} \tag{24}$$

By Assumption 3, substituting the nominal environmental dynamics

$$\begin{aligned}\widehat{\boldsymbol{\tau}}_{\mathrm{ext}} &= \widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}} \ddot{\boldsymbol{\epsilon}} + \widehat{\boldsymbol{D}}_{\mathrm{sen}} \dot{\boldsymbol{\epsilon}} + \widehat{\boldsymbol{K}}_{\mathrm{sen}} \boldsymbol{\epsilon}, \\ \int \widehat{\boldsymbol{\tau}}_{\mathrm{ext}} &= \widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}} \dot{\boldsymbol{\epsilon}} + \widehat{\boldsymbol{D}}_{\mathrm{sen}} \boldsymbol{\epsilon} + \widehat{\boldsymbol{K}}_{\mathrm{sen}} \int \boldsymbol{\epsilon}\end{aligned}$$

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into the system (24) yields the interconnected system of the Euler–Lagrange system and the environment as

$$\begin{aligned}\ddot{\boldsymbol{e}} =& -\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{\Lambda}^{-1}\boldsymbol{K}\int \boldsymbol{e} - \left(\boldsymbol{\Lambda}^{-1}\boldsymbol{K}+\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{\Lambda}^{-1}\boldsymbol{D}\right)\boldsymbol{e}\\ &-\left(\boldsymbol{\Lambda}^{-1}\boldsymbol{D}+\boldsymbol{M}^{-1}\boldsymbol{C}\right)\dot{\boldsymbol{e}} - \boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{K}}_{\mathrm{sen}}\int\boldsymbol{\epsilon}\\ &-\left(\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{K}}_{\mathrm{sen}}+\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{D}}_{\mathrm{sen}}\right)\boldsymbol{\epsilon}\\ &-\left(\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{D}}_{\mathrm{sen}}+\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\right)\dot{\boldsymbol{\epsilon}} - \boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\ddot{\boldsymbol{\epsilon}}\\ &+\boldsymbol{M}^{-1}\boldsymbol{u}+\boldsymbol{M}^{-1}\boldsymbol{w}. \end{aligned} \tag{25}$$

Inserting the impedance matching condition (23) into (25), followed by algebraic manipulation, we obtain

$$\begin{aligned}&\ddot{\boldsymbol{e}}+\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\ddot{\boldsymbol{\epsilon}}\\ &\quad=-\left(\boldsymbol{\Lambda}^{-1}\boldsymbol{D}+\boldsymbol{M}^{-1}\boldsymbol{C}\right)\left(\dot{\boldsymbol{e}}+\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\dot{\boldsymbol{\epsilon}}\right)\\ &\qquad-\left(\boldsymbol{\Lambda}^{-1}\boldsymbol{K}+\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{\Lambda}^{-1}\boldsymbol{D}\right)\left(\boldsymbol{e}+\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\boldsymbol{\epsilon}\right)\\ &\qquad-\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{\Lambda}^{-1}\boldsymbol{K}\left(\int\boldsymbol{e}+\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\int\boldsymbol{\epsilon}\right)\\ &\qquad+\boldsymbol{M}^{-1}\boldsymbol{u}+\boldsymbol{M}^{-1}\boldsymbol{w}.\end{aligned}$$

Let us define a generalized error for the interconnected system as

$$\overline{\boldsymbol{e}} := \boldsymbol{e}+\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\boldsymbol{\epsilon}, \tag{26}$$

which serves as a natural interaction coordinate representing the total error of the interconnected system. Then, letting $\boldsymbol{K}_d := \boldsymbol{\Lambda}^{-1}\boldsymbol{D}$ and $\boldsymbol{K}_p := \boldsymbol{\Lambda}^{-1}\boldsymbol{K}$, we have

$$\begin{aligned}\ddot{\overline{\boldsymbol{e}}} =& -(\boldsymbol{M}^{-1}\boldsymbol{C}+\boldsymbol{K}_d)\dot{\overline{\boldsymbol{e}}} - (\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{K}_d+\boldsymbol{K}_p)\overline{\boldsymbol{e}}\\ &-(\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{K}_p)\int\overline{\boldsymbol{e}}+\boldsymbol{M}^{-1}\boldsymbol{u}+\boldsymbol{M}^{-1}\boldsymbol{w}.\end{aligned} \tag{27}$$

The interconnected system (27) can be expressed in state space with state $\boldsymbol{x} = \begin{bmatrix}\boldsymbol{x}_1^T & \boldsymbol{x}_2^T & \boldsymbol{x}_3^T\end{bmatrix}^T = \begin{bmatrix}\int\overline{\boldsymbol{e}}^T & \overline{\boldsymbol{e}}^T & \dot{\overline{\boldsymbol{e}}}^T\end{bmatrix}^T \in \mathbb{R}^{3n}$ as

$$\dot{\boldsymbol{x}} = \boldsymbol{A}(\boldsymbol{x},t)\boldsymbol{x}+\boldsymbol{B}(\boldsymbol{x},t)\boldsymbol{u}+\boldsymbol{B}(\boldsymbol{x},t)\boldsymbol{w}, \tag{28}$$

where

$$\boldsymbol{A}=\begin{bmatrix}\boldsymbol{0} & \boldsymbol{I} & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{I}\\ -\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{K}_p & -\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{K}_d-\boldsymbol{K}_p & -\boldsymbol{M}^{-1}\boldsymbol{C}-\boldsymbol{K}_d\end{bmatrix},$$
$$\boldsymbol{B}=\begin{bmatrix}\boldsymbol{0}\\ \boldsymbol{0}\\ \boldsymbol{M}^{-1}\end{bmatrix}. \tag{29}$$

It can be observed that the interconnected system (28) has the same structure as the Euler–Lagrange system (9), except that each state variable is expressed in terms of the generalized error $\overline{\boldsymbol{e}}$ rather than the configuration error $\boldsymbol{e}$, and that the extended disturbance $\boldsymbol{w}$ includes the environmental modeling error $\widetilde{\boldsymbol{\tau}}_{\mathrm{ext}}$. Therefore, the subsequent derivation for solving the nonlinear $\mathcal{H}_\infty$ optimal control problem for the interconnected system (28) can proceed in the same manner as in Section II.

Assuming that the cost functions $\boldsymbol{h}(\boldsymbol{x},t)$ and $\boldsymbol{k}(\boldsymbol{x},t)$ in (2) satisfy $\boldsymbol{h}^T(\boldsymbol{x},t)\boldsymbol{h}(\boldsymbol{x},t) = \boldsymbol{x}^T\boldsymbol{Q}(\boldsymbol{x},t)\boldsymbol{x}$ and $\boldsymbol{k}^T(\boldsymbol{x},t)\boldsymbol{k}(\boldsymbol{x},t) = \boldsymbol{R}(\boldsymbol{x},t)$, respectively, and that the solution to the HJI equation (4) for the interconnected system (28) takes the form $V(\boldsymbol{x},t) = (1/2)\boldsymbol{x}^T\boldsymbol{P}(\boldsymbol{x},t)\boldsymbol{x}$, where $\boldsymbol{P}(\boldsymbol{x},t)$ is not an explicit function of $\boldsymbol{x}_3 = \dot{\boldsymbol{e}}$, the problem reduces to solving the DRE given in (10), for which an analytic solution has already been presented in (12).

However, this solution has three significant limitations in achieving the objectives of impedance control. First, in the referenced work [17], scalar gains were assumed to facilitate a simpler analysis. In contrast, impedance control inherently requires independent gains, typically represented by diagonal matrices (e.g., stiff in one direction and compliant in another). The applicability of the analytic solution to diagonal gains remains unclear. Second, the analytic solution (12) does not allow zero gains; however, impedance control should allow zero stiffness and/or damping. Third, the analytic solution imposes the artificial gain condition (13), which further limits the flexibility of impedance shaping.

To address these limitations, this paper proposes an analytic solution that eliminates these constraints, thereby enabling more general and flexible $\mathcal{H}_\infty$ impedance control. To begin with, the class of analytic solutions to the DRE presented in [17] is extended to diagonal gain matrices as follows (see the Appendix):

$$\begin{aligned}\boldsymbol{P}(\boldsymbol{x},t) =& \alpha\begin{bmatrix}\boldsymbol{K}_p\boldsymbol{M}\boldsymbol{K}_p & \boldsymbol{K}_p\boldsymbol{M}\boldsymbol{K}_d & \boldsymbol{K}_p\boldsymbol{M}\\ \boldsymbol{K}_d\boldsymbol{M}\boldsymbol{K}_p & \boldsymbol{K}_d\boldsymbol{M}\boldsymbol{K}_d & \boldsymbol{K}_d\boldsymbol{M}\\ \boldsymbol{M}\boldsymbol{K}_p & \boldsymbol{M}\boldsymbol{K}_d & \boldsymbol{M}\end{bmatrix}\\ &+\alpha^2\begin{bmatrix}\boldsymbol{K}_p\boldsymbol{K}_d\boldsymbol{K}_\gamma & \boldsymbol{K}_p\boldsymbol{K}_\gamma & \boldsymbol{0}\\ \boldsymbol{K}_p\boldsymbol{K}_\gamma & \boldsymbol{K}_d\boldsymbol{K}_\gamma & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0}\end{bmatrix}-\begin{bmatrix}\overline{\boldsymbol{Q}}_{12} & \overline{\boldsymbol{Q}}_{13} & \boldsymbol{0}\\ \overline{\boldsymbol{Q}}_{13} & \overline{\boldsymbol{Q}}_{23} & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{0}\end{bmatrix},\\ \boldsymbol{Q} =& \alpha^2\begin{bmatrix}\boldsymbol{K}_p^2\boldsymbol{K}_\gamma & \boldsymbol{0} & \boldsymbol{0}\\ \boldsymbol{0} & (\boldsymbol{K}_d^2-2\boldsymbol{K}_p)\boldsymbol{K}_\gamma & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{K}_\gamma\end{bmatrix}\\ &+\begin{bmatrix}\boldsymbol{0} & \overline{\boldsymbol{Q}}_{12} & \overline{\boldsymbol{Q}}_{13}\\ \overline{\boldsymbol{Q}}_{12} & 2\overline{\boldsymbol{Q}}_{13} & \overline{\boldsymbol{Q}}_{23}\\ \overline{\boldsymbol{Q}}_{13} & \overline{\boldsymbol{Q}}_{23} & \boldsymbol{0}\end{bmatrix},\\ \boldsymbol{R} =& \left(\boldsymbol{K}_\gamma+\frac{1}{\gamma^2}\boldsymbol{I}\right)^{-1},\end{aligned} \tag{30}$$

where $\boldsymbol{K}_\gamma > \boldsymbol{0}$ is a constant diagonal and positive definite matrix, $\alpha > 0$ is a constant scalar, $\gamma > 0$ is the desired $\mathcal{L}_2$-gain, and $\overline{\boldsymbol{Q}}_{(\cdot)}$ are arbitrary constant symmetric matrices.

This class of analytic solutions always satisfies the DRE (10) for any choice of the constant symmetric matrices $\overline{\boldsymbol{Q}}_{(\cdot)}$. For a given choice of $\overline{\boldsymbol{Q}}_{(\cdot)}$, if the resulting solution matrices $\boldsymbol{P}(\boldsymbol{x},t)$ and $\boldsymbol{Q}$ satisfy

$$V(\boldsymbol{x},t) = \frac{1}{2}\boldsymbol{x}^T\boldsymbol{P}(\boldsymbol{x},t)\boldsymbol{x} \geq 0,\ \boldsymbol{h}^T(\boldsymbol{x},t)\boldsymbol{h}(\boldsymbol{x},t) = \boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x} \geq 0,$$

with $V(\boldsymbol{x}^*,t) = 0$ and $\boldsymbol{h}^T(\boldsymbol{x}^*,t)\boldsymbol{h}(\boldsymbol{x}^*,t) = 0$ at the equilibrium $\boldsymbol{x}^* = \boldsymbol{0}$, then the solution matrices are well-defined [9], since $\boldsymbol{k}^T\boldsymbol{k} = \boldsymbol{R} > \boldsymbol{0}$ is positive definite. If the stronger conditions $V(\boldsymbol{x},t) > 0$ and $\boldsymbol{h}^T(\boldsymbol{x},t)\boldsymbol{h}(\boldsymbol{x},t) > 0$ hold for nonzero $\boldsymbol{x} \neq \boldsymbol{0}$, then the solution function $V(\boldsymbol{x},t)$ can also serve as a Lyapunov function, and the cost $\boldsymbol{h}^T(\boldsymbol{x},t)\boldsymbol{h}(\boldsymbol{x},t)$ penalizes the full state $\boldsymbol{x}$ (implying zero-state observability [9]).

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The simplest choice for $\overline{\boldsymbol{Q}}_{(\cdot)}$ is to set all of them to zero, i.e., $\overline{\boldsymbol{Q}}_{12}=\overline{\boldsymbol{Q}}_{13}=\overline{\boldsymbol{Q}}_{23}=\boldsymbol{0}$, which leads to

$$\boldsymbol{P}(\boldsymbol{x},t)=\alpha\cdot\begin{bmatrix}\boldsymbol{K}_p\boldsymbol{M}\boldsymbol{K}_p+\alpha\boldsymbol{K}_p\boldsymbol{K}_d\boldsymbol{K}_\gamma & \boldsymbol{K}_p\boldsymbol{M}\boldsymbol{K}_d+\alpha\boldsymbol{K}_p\boldsymbol{K}_\gamma & \boldsymbol{K}_p\boldsymbol{M}\\ \boldsymbol{K}_d\boldsymbol{M}\boldsymbol{K}_p+\alpha\boldsymbol{K}_p\boldsymbol{K}_\gamma & \boldsymbol{K}_d\boldsymbol{M}\boldsymbol{K}_d+\alpha\boldsymbol{K}_d\boldsymbol{K}_\gamma & \boldsymbol{K}_d\boldsymbol{M}\\ \boldsymbol{M}\boldsymbol{K}_p & \boldsymbol{M}\boldsymbol{K}_d & \boldsymbol{M}\end{bmatrix},$$
$$\boldsymbol{Q}=\alpha^2\begin{bmatrix}\boldsymbol{K}_p^2\boldsymbol{K}_\gamma & \boldsymbol{0} & \boldsymbol{0}\\ \boldsymbol{0} & (\boldsymbol{K}_d^2-2\boldsymbol{K}_p)\boldsymbol{K}_\gamma & \boldsymbol{0}\\ \boldsymbol{0} & \boldsymbol{0} & \boldsymbol{K}_\gamma\end{bmatrix}. \tag{31}$$

To ensure the well-definedness of the solution matrices (31), it is required that $\boldsymbol{P}(\boldsymbol{x},t)\geq\boldsymbol{0}$ and $\boldsymbol{Q}\geq\boldsymbol{0}$. First, $\boldsymbol{P}(\boldsymbol{x},t)\geq\boldsymbol{0}$ if and only if $\boldsymbol{K}_d^2\geq\boldsymbol{K}_p$, since $\boldsymbol{M}>\boldsymbol{0}$ and the Schur complement yields the condition $\boldsymbol{K}_p\boldsymbol{K}_\gamma(\boldsymbol{K}_d-\boldsymbol{K}_p\boldsymbol{K}_d^{-1})\geq\boldsymbol{0}$ for $\boldsymbol{K}_d>\boldsymbol{0}$.[4] Second, $\boldsymbol{Q}\geq\boldsymbol{0}$ if and only if $\boldsymbol{K}_d^2\geq2\boldsymbol{K}_p$. Consequently, it can be seen that an artificial gain condition $\boldsymbol{K}_d^2\geq2\boldsymbol{K}_p$ is imposed to ensure the well-definedness of both $\boldsymbol{P}(\boldsymbol{x},t)$ and $\boldsymbol{Q}$, since the latter condition is more stringent.

If the stronger conditions $\boldsymbol{P}(\boldsymbol{x},t)>\boldsymbol{0}$ and $\boldsymbol{Q}>\boldsymbol{0}$ hold, the gain condition is strengthened to $\boldsymbol{K}_d^2>2\boldsymbol{K}_p$ for $\boldsymbol{K}_d>\boldsymbol{0}$ and $\boldsymbol{K}_p>\boldsymbol{0}$. By setting $\boldsymbol{K}_d=k_d\boldsymbol{I}$ and $\boldsymbol{K}_p=k_p\boldsymbol{I}$, the solution reduces to (12) presented in [17]. Setting $\alpha=1$ leads to the solution shown in [18] and [19]. Indeed, in the stability analyses of [17]–[19], the solution function $V(\boldsymbol{x},t)>0$ is employed as a Lyapunov function together with $\boldsymbol{h}^T(\boldsymbol{x},t)\boldsymbol{h}(\boldsymbol{x},t)>0$, which necessitates the stronger conditions. It can be inferred that the gain condition implies $\boldsymbol{D}^2>2\boldsymbol{\Lambda}\boldsymbol{K}$, given that $\boldsymbol{K}_d=\boldsymbol{\Lambda}^{-1}\boldsymbol{D}$ and $\boldsymbol{K}_p=\boldsymbol{\Lambda}^{-1}\boldsymbol{K}$. Note that this solution does not allow zero stiffness and/or damping, since $\boldsymbol{K}_d>\boldsymbol{0}$ and $\boldsymbol{K}_p>\boldsymbol{0}$.

To eliminate this gain condition, the following choice for $\overline{\boldsymbol{Q}}_{(\cdot)}$ is proposed:

$$\overline{\boldsymbol{Q}}_{12}=\alpha^2\boldsymbol{K}_p\boldsymbol{K}_d\boldsymbol{K}_\gamma,\quad\overline{\boldsymbol{Q}}_{13}=\alpha^2\boldsymbol{K}_p\boldsymbol{K}_\gamma,\quad\overline{\boldsymbol{Q}}_{23}=\alpha^2\boldsymbol{K}_d\boldsymbol{K}_\gamma.$$

With this choice, the solution matrices reduce to

$$\begin{aligned}\boldsymbol{P}(\boldsymbol{x},t)&=\alpha\begin{bmatrix}\boldsymbol{K}_p\boldsymbol{M}\boldsymbol{K}_p & \boldsymbol{K}_p\boldsymbol{M}\boldsymbol{K}_d & \boldsymbol{K}_p\boldsymbol{M}\\ \boldsymbol{K}_d\boldsymbol{M}\boldsymbol{K}_p & \boldsymbol{K}_d\boldsymbol{M}\boldsymbol{K}_d & \boldsymbol{K}_d\boldsymbol{M}\\ \boldsymbol{M}\boldsymbol{K}_p & \boldsymbol{M}\boldsymbol{K}_d & \boldsymbol{M}\end{bmatrix}\\&=\alpha\begin{bmatrix}\boldsymbol{K}_p\\ \boldsymbol{K}_d\\ \boldsymbol{I}\end{bmatrix}\boldsymbol{M}\begin{bmatrix}\boldsymbol{K}_p & \boldsymbol{K}_d & \boldsymbol{I}\end{bmatrix}\end{aligned} \tag{32}$$

and

$$\begin{aligned}\boldsymbol{Q}&=\alpha^2\begin{bmatrix}\boldsymbol{K}_p^2\boldsymbol{K}_\gamma & \boldsymbol{K}_p\boldsymbol{K}_d\boldsymbol{K}_\gamma & \boldsymbol{K}_p\boldsymbol{K}_\gamma\\ \boldsymbol{K}_p\boldsymbol{K}_d\boldsymbol{K}_\gamma & \boldsymbol{K}_d^2\boldsymbol{K}_\gamma & \boldsymbol{K}_d\boldsymbol{K}_\gamma\\ \boldsymbol{K}_p\boldsymbol{K}_\gamma & \boldsymbol{K}_d\boldsymbol{K}_\gamma & \boldsymbol{K}_\gamma\end{bmatrix}\\&=\alpha^2\begin{bmatrix}\boldsymbol{K}_p\\ \boldsymbol{K}_d\\ \boldsymbol{I}\end{bmatrix}\boldsymbol{K}_\gamma\begin{bmatrix}\boldsymbol{K}_p & \boldsymbol{K}_d & \boldsymbol{I}\end{bmatrix}.\end{aligned} \tag{33}$$

Since $\boldsymbol{M}>\boldsymbol{0}$ and $\boldsymbol{K}_\gamma>\boldsymbol{0}$, it can be seen from (32) and (33) that $\boldsymbol{P}(\boldsymbol{x},t)\geq\boldsymbol{0}$ and $\boldsymbol{Q}\geq\boldsymbol{0}$ hold for all $\boldsymbol{K}_d\geq\boldsymbol{0}$ and $\boldsymbol{K}_p\geq\boldsymbol{0}$. Therefore, the solution matrices (32) and (33) are well-defined without the gain condition, and zero stiffness and/or damping are allowed.

[4]When $k_{d,i}=0$ for some axis i, the Schur complement argument is not applicable. In this case, $\boldsymbol{P}(\boldsymbol{x},t)\geq\boldsymbol{0}$ holds only if $k_{p,i}=0$ for all such i; for example, $\boldsymbol{K}_d=\boldsymbol{0}$ and $\boldsymbol{K}_p=\boldsymbol{0}$ trivially yield $\boldsymbol{P}(\boldsymbol{x},t)\geq\boldsymbol{0}$.

Moreover, let us define an auxiliary variable $\boldsymbol{s}\in\mathbb{R}^n$ as

$$\boldsymbol{s}:=\begin{bmatrix}\boldsymbol{K}_p & \boldsymbol{K}_d & \boldsymbol{I}\end{bmatrix}\boldsymbol{x}=\dot{\overline{\boldsymbol{e}}}+\boldsymbol{K}_d\overline{\boldsymbol{e}}+\boldsymbol{K}_p\int\overline{\boldsymbol{e}}. \tag{34}$$

Then,

$$V(\boldsymbol{x},t)=\frac{1}{2}\boldsymbol{x}^T\boldsymbol{P}(\boldsymbol{x},t)\boldsymbol{x}=\frac{1}{2}\alpha\boldsymbol{s}^T\boldsymbol{M}\boldsymbol{s}=V(\boldsymbol{s},t)>0 \tag{35}$$

and

$$\boldsymbol{h}^T(\boldsymbol{x},t)\boldsymbol{h}(\boldsymbol{x},t)=\boldsymbol{x}^T\boldsymbol{Q}\boldsymbol{x}=\alpha^2\boldsymbol{s}^T\boldsymbol{K}_\gamma\boldsymbol{s}=\boldsymbol{h}^T(\boldsymbol{s},t)\boldsymbol{h}(\boldsymbol{s},t)>0$$

hold for nonzero $\boldsymbol{s}\neq\boldsymbol{0}$, and vanish on $\boldsymbol{s}=\boldsymbol{0}$. Therefore, $V(\boldsymbol{s},t)$ can serve as a Lyapunov function with respect to $\boldsymbol{s}$.

In fact, the proposed $\mathcal{H}_\infty$ impedance control, as well as the $\mathcal{H}_\infty$ optimal control laws in [17]–[19], operates by driving the auxiliary variable $\boldsymbol{s}$, rather than the full state $\boldsymbol{x}$ directly, to zero, as will be proven in Section IV. This can be interpreted as the auxiliary variable $\boldsymbol{s}$ playing a role akin to that of the sliding variable in SMC [5], where $\boldsymbol{s}=\boldsymbol{0}$ defines the sliding surface. Although the proposed framework and those in [17]–[19] are explicitly derived from the nonlinear $\mathcal{H}_\infty$ optimal control problem, their behavior resembles that of SMC.

With the proposed solution matrices (32) and (33), the nonlinear $\mathcal{H}_\infty$ optimal control that satisfies the $\mathcal{L}_2$-gain attenuation requirement (11) for the interconnected system (28) is given by

$$\begin{aligned}\boldsymbol{u}&=-\left(\boldsymbol{K}_\gamma+\frac{1}{\gamma^2}\boldsymbol{I}\right)\boldsymbol{B}^T(\boldsymbol{x},t)\boldsymbol{P}(\boldsymbol{x},t)\boldsymbol{x}\\&=-\alpha\left(\boldsymbol{K}_\gamma+\frac{1}{\gamma^2}\boldsymbol{I}\right)\left(\dot{\overline{\boldsymbol{e}}}+\boldsymbol{K}_d\overline{\boldsymbol{e}}+\boldsymbol{K}_p\int\overline{\boldsymbol{e}}\right)\\&=-\alpha\left(\boldsymbol{K}_\gamma+\frac{1}{\gamma^2}\boldsymbol{I}\right)\boldsymbol{s}.\end{aligned} \tag{36}$$

As the auxiliary variable $\boldsymbol{s}$ converges to zero, the optimal control input $\boldsymbol{u}$ likewise becomes zero. One notable observation is that the resulting optimal control $\boldsymbol{u}$ is invariant to the specific choices of $\overline{\boldsymbol{Q}}_{(\cdot)}$, since the last term of $\boldsymbol{P}$ in (30) is annihilated by the zero blocks of $\boldsymbol{B}$ in (29). This implies that the nonlinear $\mathcal{H}_\infty$ optimal control laws in [17]–[19] can also be formulated without the gain condition.

Substituting the generalized error (26), the gain relationships $\boldsymbol{K}_p=\boldsymbol{\Lambda}^{-1}\boldsymbol{K}$ and $\boldsymbol{K}_d=\boldsymbol{\Lambda}^{-1}\boldsymbol{D}$, and the impedance matching condition (23) into the auxiliary variable $\boldsymbol{s}$ (34) yields

$$\begin{aligned}\boldsymbol{s}&=\left(\dot{\boldsymbol{e}}+\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\dot{\boldsymbol{\epsilon}}\right)+\boldsymbol{\Lambda}^{-1}\boldsymbol{D}\left(\boldsymbol{e}+\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\boldsymbol{\epsilon}\right)\\&\quad+\boldsymbol{\Lambda}^{-1}\boldsymbol{K}\left(\int\boldsymbol{e}+\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\int\boldsymbol{\epsilon}\right)\\&=\dot{\boldsymbol{e}}+\boldsymbol{\Lambda}^{-1}\boldsymbol{D}\boldsymbol{e}+\boldsymbol{\Lambda}^{-1}\boldsymbol{K}\int\boldsymbol{e}\\&\quad+\boldsymbol{\Lambda}^{-1}\left(\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\dot{\boldsymbol{\epsilon}}+\boldsymbol{D}\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\boldsymbol{\epsilon}+\boldsymbol{K}\boldsymbol{\Lambda}^{-1}\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\int\boldsymbol{\epsilon}\right)\\&=\dot{\boldsymbol{e}}+\boldsymbol{\Lambda}^{-1}\boldsymbol{D}\boldsymbol{e}+\boldsymbol{\Lambda}^{-1}\boldsymbol{K}\int\boldsymbol{e}\\&\quad+\boldsymbol{\Lambda}^{-1}\left(\widehat{\boldsymbol{\Lambda}}_{\mathrm{sen}}\dot{\boldsymbol{\epsilon}}+\widehat{\boldsymbol{D}}_{\mathrm{sen}}\boldsymbol{\epsilon}+\widehat{\boldsymbol{K}}_{\mathrm{sen}}\int\boldsymbol{\epsilon}\right)\\&=\dot{\boldsymbol{e}}+\boldsymbol{\Lambda}^{-1}\boldsymbol{D}\boldsymbol{e}+\boldsymbol{\Lambda}^{-1}\boldsymbol{K}\int\boldsymbol{e}+\boldsymbol{\Lambda}^{-1}\int\widehat{\boldsymbol{\tau}}_{\mathrm{ext}}\\&=\dot{\boldsymbol{e}}_{\mathrm{ref}}.\end{aligned} \tag{37}$$

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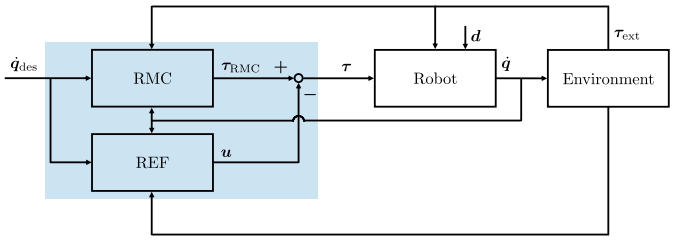


Fig. 2. Block diagram of the proposed $\mathcal{H}_\infty$ impedance control. The blue-shaded box represents the proposed $\mathcal{H}_\infty$ impedance control via RMC with REF.

Finally, the resulting optimal control is given by

$$\boldsymbol{u} = -\alpha \left(\boldsymbol{K}_\gamma + \frac{1}{\gamma^2} \boldsymbol{I} \right) \dot{\boldsymbol{e}}_{\rm ref}, \tag{38}$$

which takes the form of the REF. □

Remark 2: Although the nonlinear $\mathcal{H}_\infty$ optimal control (36) is derived for the interconnected system (28), in which the environmental dynamics is explicitly considered, all environmental parameters are completely canceled out within the final control law (38), making their explicit specification unnecessary. Consequently, no identification of the environment is required, nor is it necessary to measure or estimate the displacements $\boldsymbol{\delta}$ and $\boldsymbol{\epsilon}$ and their derivatives. These desirable properties of the proposed controller allow it to generalize to any environment satisfying Assumption 2.

Remark 3: The proposed analytic solution, given by (32) and (33), eliminates the gain condition $\boldsymbol{K}_d^2 > 2\boldsymbol{K}_p$. Since it does not alter the resulting optimal control, the gain condition can also be eliminated in the formulations presented in [17]–[19].

Substituting (38) into (18) yields the final form of the proposed $\mathcal{H}_\infty$ impedance control via RMC with REF as

$$\boldsymbol{\tau} = \widehat{\boldsymbol{M}} \ddot{\boldsymbol{q}}_{\rm ref} + \widehat{\boldsymbol{C}} \dot{\boldsymbol{q}}_{\rm ref} + \widehat{\boldsymbol{g}} - \widehat{\boldsymbol{\tau}}_{\rm ext} + \alpha \left(\boldsymbol{K}_\gamma + \frac{1}{\gamma^2} \boldsymbol{I} \right) \dot{\boldsymbol{e}}_{\rm ref}, \tag{39}$$

where the dependence on $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ is omitted for brevity. The block diagram of the proposed $\mathcal{H}_\infty$ impedance control via RMC with REF (39) is illustrated in Fig. 2. Notably, it neither generates a motion command nor tracks one, and it clearly exhibits an impedance control structure with the opposite causality to that of the $\mathcal{H}_\infty$ impedance control via the NRIC framework presented in Fig. 1.

IV. Stability Analysis

Definition 1: A continuous function $\kappa : [0, a) \to \mathbb{R}_+$ is said to belong to class $\mathcal{K}$ if it is strictly increasing and $\kappa(0) = 0$. The function is said to belong to class $\mathcal{K}_\infty$ if $a = \infty$ and $\kappa(r) \to \infty$ as $r \to \infty$. A continuous function $\rho : [0, a) \times \mathbb{R}_+ \to \mathbb{R}_+$ is said to belong to class $\mathcal{KL}$ if, for each fixed s, the mapping $\rho(\cdot, s)$ belongs to class $\mathcal{K}$, and, for each fixed r, $\rho(r, \cdot)$ is decreasing and satisfies $\rho(r, s) \to 0$ as $s \to \infty$.

Definition 2: The extended disturbance-affine system

$$\dot{\boldsymbol{s}} = \boldsymbol{f}(\boldsymbol{s}, t) + \boldsymbol{g}(\boldsymbol{s}, t)\boldsymbol{w} \tag{40}$$

is said to be extended disturbance ISS if there exist a class $\mathcal{KL}$ function ρ and a class $\mathcal{K}$ function κ such that the solution of (40) exists for all $t \geq 0$ and satisfies

$$||\boldsymbol{s}(t)|| \leq \rho(||\boldsymbol{s}(0)||, t) + \kappa \left(\sup_{0 \leq \sigma \leq t} ||\boldsymbol{w}(\sigma)|| \right)$$

for every initial state $\boldsymbol{s}(0)$ and every extended disturbance $\boldsymbol{w}(\cdot)$ that is piecewise continuous and bounded on $[0, \infty)$.

Equivalently, ISS can be characterized by means of a Lyapunov function. The system (40) is extended disturbance ISS if and only if there exist a smooth positive definite radially unbounded function $V(\boldsymbol{s}, t)$, a class $\mathcal{K}_\infty$ function κ_1, and a class $\mathcal{K}$ function κ_2 such that

$$\dot{V} \leq -\kappa_1(||\boldsymbol{s}||) + \kappa_2(||\boldsymbol{w}||),$$

where $\dot{V}$ denotes the total derivative of the Lyapunov function.

Remark 4: The system (28) can be expressed in terms of the auxiliary variable $\boldsymbol{s}$ (34) as

$$\dot{\boldsymbol{s}} = \boldsymbol{A}'(\boldsymbol{s}, t)\boldsymbol{s} + \boldsymbol{B}'(\boldsymbol{s}, t)\boldsymbol{u} + \boldsymbol{B}'(\boldsymbol{s}, t)\boldsymbol{w}, \tag{41}$$

where $\boldsymbol{A}' = -\boldsymbol{M}^{-1}\boldsymbol{C}$ and $\boldsymbol{B}' = \boldsymbol{M}^{-1}$.

Theorem 2: If the $\mathcal{H}_\infty$ impedance control (38) is applied to the interconnected system (28), then the system is extended disturbance ISS with respect to the auxiliary variable $\boldsymbol{s}$ (34). Consequently, the $\mathcal{H}_\infty$ impedance control achieves robust stability and performance in rendering the desired impedance under bounded extended disturbance.

Proof: Define the Lyapunov function for the interconnected system (28) as follows:

$$V(\boldsymbol{x}, t) = \frac{1}{2} \boldsymbol{x}^T \boldsymbol{P}(\boldsymbol{x}, t) \boldsymbol{x},$$

with the state $\boldsymbol{x} = \begin{bmatrix} \int \overline{\boldsymbol{e}}^T & \overline{\boldsymbol{e}}^T & \dot{\overline{\boldsymbol{e}}}^T \end{bmatrix}^T \in \mathbb{R}^{3n}$ and the solution matrix $\boldsymbol{P}(\boldsymbol{x}, t)$ in (32).

First, the Lyapunov function V is differentiable and positive definite with respect to the auxiliary variable $\boldsymbol{s}$ by (35).

Second, the Lyapunov function V is radially unbounded with respect to $\boldsymbol{s}$ since $V(\boldsymbol{s}, t) \to \infty$ as $\boldsymbol{s} \to \infty$ by (35).

Third, the total derivative of the Lyapunov function V along the trajectories of (28) is given by

$$\dot{V} = \underbrace{V_t + \boldsymbol{V}_x \boldsymbol{A}\boldsymbol{x}}_{\rm (i)} + \underbrace{\boldsymbol{V}_x \boldsymbol{B}\boldsymbol{u} + \boldsymbol{V}_x \boldsymbol{B}\boldsymbol{w}}_{\rm (ii)},$$

and its components are calculated as

$$\begin{aligned} {\rm (i)} &= \frac{1}{2} \boldsymbol{x}^T \left(\dot{\boldsymbol{P}} + \boldsymbol{P}\boldsymbol{A} + \boldsymbol{A}^T \boldsymbol{P} \right) \boldsymbol{x} \\ &= \frac{1}{2} \alpha \boldsymbol{s}^T \left(\dot{\boldsymbol{M}} - \boldsymbol{C} - \boldsymbol{C}^T \right) \boldsymbol{s} \\ &= 0, \\ {\rm (ii)} &= \boldsymbol{x}^T \boldsymbol{P}\boldsymbol{B}\boldsymbol{u} + \boldsymbol{x}^T \boldsymbol{P}\boldsymbol{B}\boldsymbol{w} \\ &= \alpha \boldsymbol{s}^T \boldsymbol{u} + \alpha \boldsymbol{s}^T \boldsymbol{w}. \end{aligned} \tag{42}$$

Substituting the optimal control $\boldsymbol{u}$ (36) into (42) yields

$$\begin{aligned} \dot{V} &= -\alpha^2 \boldsymbol{s}^T \left(\boldsymbol{K}_\gamma + \frac{1}{\gamma^2} \boldsymbol{I} \right) \boldsymbol{s} + \alpha \boldsymbol{s}^T \boldsymbol{w} \\ &\leq -\alpha^2 \beta \boldsymbol{s}^T \boldsymbol{s} + \alpha \boldsymbol{s}^T \boldsymbol{w}, \end{aligned}$$

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where $\beta := \lambda_{\min}(\boldsymbol{K}_\gamma) + 1/\gamma^2$ and $\lambda_{\min}(\cdot)$ denotes the minimum eigenvalue of $(\cdot)$. Applying Young's inequality to the cross term gives $\boldsymbol{s}^T\boldsymbol{w} \le (\alpha\beta/2)||\boldsymbol{s}||^2 + (1/(2\alpha\beta))||\boldsymbol{w}||^2$. Therefore, $\dot{V}$ can be further upper bounded as

$$\begin{aligned}\dot{V} &\le -\frac{\alpha^2\beta}{2}||\boldsymbol{s}||^2 + \frac{1}{2\beta}||\boldsymbol{w}||^2 \\ &=: -\kappa_1(||\boldsymbol{s}||) + \kappa_2(||\boldsymbol{w}||), \end{aligned} \tag{43}$$

where $\kappa_1 \in \mathcal{K}_\infty$ and $\kappa_2 \in \mathcal{K}$. By Definition 2, it is concluded that the interconnected system (28) is extended disturbance ISS with respect to the auxiliary variable $\boldsymbol{s}$, provided that the $\mathcal{H}_\infty$ impedance control is applied.

It can be observed that $\dot{V} < 0$ holds for $||\boldsymbol{s}|| > ||\boldsymbol{w}||/(\alpha\beta)$; hence, the ultimate bound on $\boldsymbol{s}$ is inversely proportional to $\alpha\beta$. In particular, as the $\mathcal{L}_2$-gain γ decreases, the detrimental term $(1/2\beta)||\boldsymbol{w}||^2$ becomes less effective, scaling with γ^2, whereas the beneficial term $(\alpha^2\beta/2)||\boldsymbol{s}||^2$ increases proportionally to $1/\gamma^2$. The $\mathcal{L}_2$-gain $\gamma > 0$ can be decreased until oscillations occur due to hardware limitations, and it can be easily tuned using the *square-linear tuning* rule [18], [19].

If the condition

$$||\boldsymbol{w}|| \le \theta\alpha\beta||\boldsymbol{s}||, \qquad 0 < \theta < 1 \tag{44}$$

holds for all $\boldsymbol{s}$ and $\boldsymbol{w}$, then $\dot{V} \le -(1-\theta)\alpha^2\beta||\boldsymbol{s}||^2$, and the system is not only ISS but even globally asymptotically stable. That is, the auxiliary variable $\boldsymbol{s}$ converges to zero. Since the auxiliary variable $\boldsymbol{s}$ is equal to the reference error $\dot{\boldsymbol{e}}_{\rm ref}$ as derived in (37), the $\mathcal{H}_\infty$ impedance control (38) inherently aims to drive $\dot{\boldsymbol{e}}_{\rm ref} \to \boldsymbol{0}$ under bounded extended disturbance. This convergence leads to $\ddot{\boldsymbol{e}}_{\rm ref} \to \boldsymbol{0}$, implying that the desired impedance is effectively rendered. Indeed, as $\boldsymbol{s}$ converges to zero, both $\boldsymbol{u}$ and $\boldsymbol{w}$ vanish by (36) and (44); hence, by (41), $\dot{\boldsymbol{s}} = \ddot{\boldsymbol{e}}_{\rm ref} \to \boldsymbol{0}$. □

Corollary 1: If the $\mathcal{H}_\infty$ impedance control (38) is applied to the interconnected system (28), the following hold:

(i) If (44) holds, then:
  (a) For $\boldsymbol{K}_d > \boldsymbol{0}$ and $\boldsymbol{K}_p > \boldsymbol{0}$, the state $\boldsymbol{x}$ converges to the equilibrium $\boldsymbol{x}^* = \boldsymbol{0}$.
  (b) For $\boldsymbol{K}_d = \boldsymbol{0}$ or $\boldsymbol{K}_p = \boldsymbol{0}$, but not both, the state $\boldsymbol{x}$ remains bounded but does not converge to the equilibrium in general.
  (c) For $\boldsymbol{K}_d = \boldsymbol{0}$ and $\boldsymbol{K}_p = \boldsymbol{0}$, the state $\boldsymbol{x}$ is not guaranteed to remain bounded in general.

(ii) Otherwise:
  (a) For $\boldsymbol{K}_d > \boldsymbol{0}$ and $\boldsymbol{K}_p > \boldsymbol{0}$, the state $\boldsymbol{x}$ remains bounded.
  (b) For $\boldsymbol{K}_d = \boldsymbol{0}$ or $\boldsymbol{K}_p = \boldsymbol{0}$, the state $\boldsymbol{x}$ is not guaranteed to remain bounded in general.

Proof: (i) If (44) holds, the auxiliary variable $\boldsymbol{s}$ converges to zero by Theorem 2, i.e., the system trajectories converge to the manifold $\boldsymbol{s} = \boldsymbol{0}$, analogous to the sliding manifold in SMC, on which the behavior is governed by the reduced-order dynamics

$$\boldsymbol{0} = \dot{\overline{\boldsymbol{e}}} + \boldsymbol{K}_d\overline{\boldsymbol{e}} + \boldsymbol{K}_p\int\overline{\boldsymbol{e}}.$$

For $\boldsymbol{K}_d > \boldsymbol{0}$ and $\boldsymbol{K}_p > \boldsymbol{0}$, the system state $\boldsymbol{x}$ converges to the equilibrium $\boldsymbol{x}^* = \boldsymbol{0}$. For $\boldsymbol{K}_d > \boldsymbol{0}$ and $\boldsymbol{K}_p = \boldsymbol{0}$ (zero stiffness), $\dot{\overline{\boldsymbol{e}}}$ and $\overline{\boldsymbol{e}}$ converge to zero, while $\int\overline{\boldsymbol{e}}$ remains bounded. For $\boldsymbol{K}_d = \boldsymbol{0}$ and $\boldsymbol{K}_p > \boldsymbol{0}$ (zero damping), the state remains bounded and generally oscillates without converging to the equilibrium. For $\boldsymbol{K}_d = \boldsymbol{0}$ and $\boldsymbol{K}_p = \boldsymbol{0}$ (zero damping and stiffness), $\dot{\overline{\boldsymbol{e}}}$ converges to zero, while $\overline{\boldsymbol{e}}$ remains constant and nonzero in general, so that $\int\overline{\boldsymbol{e}}$ may drift.

(ii) By Theorem 2, the auxiliary variable $\boldsymbol{s}$ remains bounded. For $\boldsymbol{K}_d > \boldsymbol{0}$ and $\boldsymbol{K}_p > \boldsymbol{0}$, the state remains bounded by (34). For $\boldsymbol{K}_d > \boldsymbol{0}$ and $\boldsymbol{K}_p = \boldsymbol{0}$, $\dot{\overline{\boldsymbol{e}}}$ and $\overline{\boldsymbol{e}}$ remain bounded, while $\int\overline{\boldsymbol{e}}$ may drift. For $\boldsymbol{K}_d = \boldsymbol{0}$ and $\boldsymbol{K}_p \ge \boldsymbol{0}$, the boundedness of the state is not guaranteed in general. □

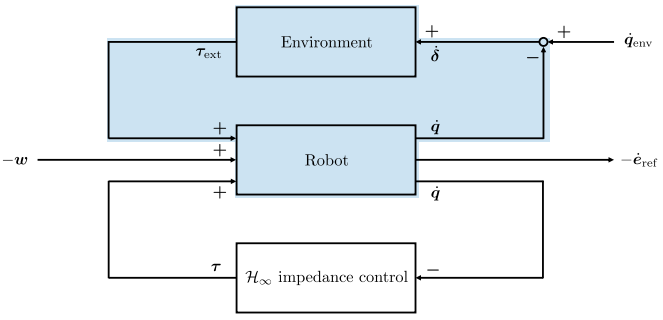


Fig. 3. Feedback interconnection between the robot and the environment. The blue-shaded region represents the interconnected robot–environment system.

Remark 5: For $\boldsymbol{K}_d = \boldsymbol{0}$ or $\boldsymbol{K}_p = \boldsymbol{0}$ (zero damping or stiffness), the lack of boundedness of the state $\boldsymbol{x}$ is inherent to the specified desired dynamics rather than a limitation of the proposed controller. In all cases of Corollary 1, the $\mathcal{L}_2$-gain attenuation requirement (11) and the ISS (43) with respect to the auxiliary variable $\boldsymbol{s}$ remain valid, which is consistent with the objective of the proposed controller: rendering the desired dynamics by driving the auxiliary variable $\boldsymbol{s}$ to zero, rather than driving the full state $\boldsymbol{x}$. Note that only $\boldsymbol{s}$, not $\boldsymbol{x}$, appears in the control input $\boldsymbol{u}$ (36); hence, $\boldsymbol{u}$ remains bounded.

Remark 6: For $\boldsymbol{K}_d > \boldsymbol{0}$ and $\boldsymbol{K}_p > \boldsymbol{0}$, the proposed $\mathcal{H}_\infty$ impedance control (38) stabilizes the interconnected system (28). Consequently, the Euler–Lagrange system (5) and the environment (21) remain bounded. Indeed, the boundedness of the state $\boldsymbol{x}$ implies those of $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\boldsymbol{\epsilon}$, and $\dot{\boldsymbol{\epsilon}}$ by Assumption 3 and the impedance matching condition (23), and the boundedness of $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ in turn implies those of $\boldsymbol{\delta}$ and $\dot{\boldsymbol{\delta}}$ by Assumption 2. For $\boldsymbol{K}_d > \boldsymbol{0}$ and $\boldsymbol{K}_p = \boldsymbol{0}$, the Euler–Lagrange system and the environment remain bounded (in the sense that $\boldsymbol{q}$, $\dot{\boldsymbol{q}}$, $\boldsymbol{\delta}$, and $\dot{\boldsymbol{\delta}}$ remain bounded) provided that $\int\widehat{\boldsymbol{\tau}}_{\rm ext}$ remains bounded; otherwise, e.g., under a sustained unidirectional interaction force, $\boldsymbol{q}$ and $\boldsymbol{\delta}$ may drift, while $\dot{\boldsymbol{q}}$ and $\dot{\boldsymbol{\delta}}$ remain bounded. For $\boldsymbol{K}_d = \boldsymbol{0}$ and $\boldsymbol{K}_p \ge \boldsymbol{0}$, the boundedness of the Euler–Lagrange system and the environment cannot be guaranteed in general; however, this is a natural consequence of the undamped target dynamics under the disturbance input.

V. Passivity Analysis

Definition 3: A system is passive if there exists a continuously differentiable scalar function $S(\boldsymbol{x}) \ge 0$, called a storage function, such that the following inequality holds:

$$\dot{S}(\boldsymbol{x}) \le \boldsymbol{u}^T\boldsymbol{y},$$

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where $\boldsymbol{x}$, $\boldsymbol{u}$, and $\boldsymbol{y}$ represent the state, input, and output of the system, respectively.

Theorem 3: For the $\mathcal{H}_\infty$ impedance-controlled robot (19) that physically interacts with the environment (21) through the interaction port $(\boldsymbol{\tau}_{\rm ext}, \dot{\boldsymbol{q}})$, as illustrated in Fig. 3, the robot operates as a mechanical admittance with respect to the input–output pair

$$\left(\begin{bmatrix} -\boldsymbol{w} \\ \boldsymbol{\tau} \\ \boldsymbol{\tau}_{\rm ext} \end{bmatrix}, \begin{bmatrix} -\dot{\boldsymbol{e}}_{\rm ref} \\ \dot{\boldsymbol{q}} \\ \dot{\boldsymbol{q}} \end{bmatrix} \right), \tag{45}$$

whereas the environment behaves as a mechanical impedance with respect to

$$\left(\dot{\boldsymbol{\delta}}, \boldsymbol{\tau}_{\rm ext} \right). \tag{46}$$

The robot is passive with respect to the input–output pair (45), the environment is passive with respect to (46), and consequently, the interconnected robot–environment system is passive with respect to

$$\left(\begin{bmatrix} -\boldsymbol{w} \\ \boldsymbol{\tau} \\ \dot{\boldsymbol{q}}_{\rm env} \end{bmatrix}, \begin{bmatrix} -\dot{\boldsymbol{e}}_{\rm ref} \\ \dot{\boldsymbol{q}} \\ \boldsymbol{\tau}_{\rm ext} \end{bmatrix} \right). \tag{47}$$

Proof: First, we show that the $\mathcal{H}_\infty$ impedance-controlled robot (19) is passive with respect to the input–output pair (45). Let us define the storage function $S_{\rm robot}$ as follows:

$$S_{\rm robot} = \frac{1}{2}\dot{\boldsymbol{e}}_{\rm ref}^T \boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{e}}_{\rm ref} + \frac{1}{2}\dot{\boldsymbol{q}}^T \boldsymbol{M}(\boldsymbol{q})\dot{\boldsymbol{q}} + U(\boldsymbol{q}) \geq 0,$$

where $U(\boldsymbol{q}) \geq 0$ denotes the gravitational potential energy function such that $\partial U(\boldsymbol{q})/\partial \boldsymbol{q} = \boldsymbol{g}(\boldsymbol{q})$. The time derivative of the storage function $\dot{S}_{\rm robot}$ can be calculated as follows:

$$\dot{S}_{\rm robot} = \underbrace{\dot{\boldsymbol{e}}_{\rm ref}^T \boldsymbol{M}\ddot{\boldsymbol{e}}_{\rm ref} + \frac{1}{2}\dot{\boldsymbol{e}}_{\rm ref}^T \dot{\boldsymbol{M}}\dot{\boldsymbol{e}}_{\rm ref}}_{\rm (i)} + \underbrace{\dot{\boldsymbol{q}}^T \boldsymbol{M}\ddot{\boldsymbol{q}} + \frac{1}{2}\dot{\boldsymbol{q}}^T \dot{\boldsymbol{M}}\dot{\boldsymbol{q}} + \dot{\boldsymbol{q}}^T \boldsymbol{g}}_{\rm (ii)}.$$

From the $\mathcal{H}_\infty$ impedance-controlled robot (19), substituting $\boldsymbol{M}\ddot{\boldsymbol{e}}_{\rm ref} = -\boldsymbol{C}\dot{\boldsymbol{e}}_{\rm ref} + \boldsymbol{u} + \boldsymbol{w}$ into (i) yields

$$\begin{aligned} {\rm (i)} &= \dot{\boldsymbol{e}}_{\rm ref}^T(-\boldsymbol{C}\dot{\boldsymbol{e}}_{\rm ref} + \boldsymbol{u} + \boldsymbol{w}) + \frac{1}{2}\dot{\boldsymbol{e}}_{\rm ref}^T \dot{\boldsymbol{M}}\dot{\boldsymbol{e}}_{\rm ref} \\ &= \frac{1}{2}\dot{\boldsymbol{e}}_{\rm ref}^T(\dot{\boldsymbol{M}} - 2\boldsymbol{C})\dot{\boldsymbol{e}}_{\rm ref} + \dot{\boldsymbol{e}}_{\rm ref}^T \boldsymbol{u} + \dot{\boldsymbol{e}}_{\rm ref}^T \boldsymbol{w} \\ &= -\alpha \dot{\boldsymbol{e}}_{\rm ref}^T \left(\boldsymbol{K}_\gamma + \frac{1}{\gamma^2}\boldsymbol{I} \right) \dot{\boldsymbol{e}}_{\rm ref} + \dot{\boldsymbol{e}}_{\rm ref}^T \boldsymbol{w} \\ &\leq \dot{\boldsymbol{e}}_{\rm ref}^T \boldsymbol{w}. \end{aligned}$$

From the third line, it can be observed that the $\mathcal{H}_\infty$ optimal control input $\boldsymbol{u}$ (38) introduces a highly dissipative effect with respect to the reference error (rather than the velocity $\dot{\boldsymbol{q}}$ or the velocity error $\dot{\boldsymbol{e}}$), enabling energy dissipation without increasing the damping and even in the absence of motion. This desirable property of the proposed controller enables stable interaction even under low-damping settings. Note that, as discussed in Section IV, $\boldsymbol{K}_\gamma + (1/\gamma^2)\boldsymbol{I}$ is tuned to be large.

From (19), substituting $\boldsymbol{M}\ddot{\boldsymbol{q}} = \boldsymbol{M}\ddot{\boldsymbol{q}}_{\rm ref} + \boldsymbol{C}\dot{\boldsymbol{q}}_{\rm ref} - \boldsymbol{C}\dot{\boldsymbol{q}} - \boldsymbol{u} - \boldsymbol{w}$ into (ii) yields

$$\begin{aligned} {\rm (ii)} &= \dot{\boldsymbol{q}}^T(\boldsymbol{M}\ddot{\boldsymbol{q}}_{\rm ref} + \boldsymbol{C}\dot{\boldsymbol{q}}_{\rm ref} - \boldsymbol{C}\dot{\boldsymbol{q}} - \boldsymbol{u} - \boldsymbol{w}) + \frac{1}{2}\dot{\boldsymbol{q}}^T \dot{\boldsymbol{M}}\dot{\boldsymbol{q}} + \dot{\boldsymbol{q}}^T \boldsymbol{g} \\ &= \frac{1}{2}\dot{\boldsymbol{q}}^T(\dot{\boldsymbol{M}} - 2\boldsymbol{C})\dot{\boldsymbol{q}} + \dot{\boldsymbol{q}}^T(\boldsymbol{M}\ddot{\boldsymbol{q}}_{\rm ref} + \boldsymbol{C}\dot{\boldsymbol{q}}_{\rm ref} + \boldsymbol{g} - \boldsymbol{u} - \boldsymbol{w}), \end{aligned}$$

where the extended disturbance $\boldsymbol{w}$ (20) can be rearranged as

$$\boldsymbol{w} = \boldsymbol{M}\ddot{\boldsymbol{q}}_{\rm ref} + \boldsymbol{C}\dot{\boldsymbol{q}}_{\rm ref} + \boldsymbol{g} + \boldsymbol{d} - \boldsymbol{\tau}_{\rm ext} - \boldsymbol{\tau}_{\rm RMC},$$

in which $\boldsymbol{\tau}_{\rm RMC} = \widehat{\boldsymbol{M}}\ddot{\boldsymbol{q}}_{\rm ref} + \widehat{\boldsymbol{C}}\dot{\boldsymbol{q}}_{\rm ref} + \widehat{\boldsymbol{g}} - \widehat{\boldsymbol{\tau}}_{\rm ext}$. Then,

$$\begin{aligned} {\rm (ii)} &= \dot{\boldsymbol{q}}^T \boldsymbol{\tau}_{\rm ext} + \dot{\boldsymbol{q}}^T(\boldsymbol{\tau}_{\rm RMC} - \boldsymbol{u}) - \dot{\boldsymbol{q}}^T \boldsymbol{d} \\ &\leq \dot{\boldsymbol{q}}^T \boldsymbol{\tau}_{\rm ext} + \dot{\boldsymbol{q}}^T \boldsymbol{\tau} \end{aligned}$$

since $-\dot{\boldsymbol{q}}^T \boldsymbol{d} \leq 0$ by Assumption 1, and $\boldsymbol{\tau} = \boldsymbol{\tau}_{\rm RMC} - \boldsymbol{u}$. Hence,

$$\begin{aligned} \dot{S}_{\rm robot} &= {\rm (i)} + {\rm (ii)} \\ &\leq (-\boldsymbol{w})^T(-\dot{\boldsymbol{e}}_{\rm ref}) + \boldsymbol{\tau}_{\rm ext}^T \dot{\boldsymbol{q}} + \boldsymbol{\tau}^T \dot{\boldsymbol{q}}. \end{aligned}$$

It is concluded that the $\mathcal{H}_\infty$ impedance-controlled robot is passive with respect to the input–output pair (45).

Second, we show that the environment (21) is passive with respect to the input–output pair (46). Let us define the storage function $S_{\rm env}$ as follows:

$$S_{\rm env} = \frac{1}{2}\dot{\boldsymbol{\delta}}^T \boldsymbol{\Lambda}_{\rm env}\dot{\boldsymbol{\delta}} + \frac{1}{2}\boldsymbol{\delta}^T \boldsymbol{K}_{\rm env}\boldsymbol{\delta} \geq 0.$$

Taking the time derivative and using (21), we obtain

$$\dot{S}_{\rm env} \leq \dot{\boldsymbol{\delta}}^T \boldsymbol{\tau}_{\rm ext} = (\dot{\boldsymbol{q}}_{\rm env} - \dot{\boldsymbol{q}})^T \boldsymbol{\tau}_{\rm ext}.$$

The environment is passive with respect to (46).

Third, the storage function of the interconnected robot–environment system can be defined by $S = S_{\rm robot} + S_{\rm env}$. Then, the time derivative of the storage function S satisfies

$$\begin{aligned} \dot{S} &= \dot{S}_{\rm robot} + \dot{S}_{\rm env} \\ &\leq (-\boldsymbol{w})^T(-\dot{\boldsymbol{e}}_{\rm ref}) + \boldsymbol{\tau}^T \dot{\boldsymbol{q}} + (\boldsymbol{\tau}_{\rm ext}^T \dot{\boldsymbol{q}} - \dot{\boldsymbol{q}}^T \boldsymbol{\tau}_{\rm ext}) + \dot{\boldsymbol{q}}_{\rm env}^T \boldsymbol{\tau}_{\rm ext} \\ &= (-\boldsymbol{w})^T(-\dot{\boldsymbol{e}}_{\rm ref}) + \boldsymbol{\tau}^T \dot{\boldsymbol{q}} + \dot{\boldsymbol{q}}_{\rm env}^T \boldsymbol{\tau}_{\rm ext}. \end{aligned}$$

Consequently, it is concluded that the interconnected robot–environment system is passive with respect to the input–output pair (47), provided that the $\mathcal{H}_\infty$ impedance control is applied. □

VI. $\mathcal{H}_\infty$ Impedance Control in Operational Space

It is worth noting that impedance control is primarily designed and used in the operational (or task) space. The kinematic relationship between the end-effector (or tool) of the robot and the joint configuration is described by

$$\boldsymbol{p} = \boldsymbol{f}(\boldsymbol{q}), \qquad \dot{\boldsymbol{p}} = \boldsymbol{J}(\boldsymbol{q})\dot{\boldsymbol{q}}, \tag{48}$$

where $\boldsymbol{p} \in \mathbb{R}^m$ denotes the task coordinates of the end-effector, $\boldsymbol{f} : \mathbb{R}^n \to \mathbb{R}^m$ stands for the forward kinematic map, and $\boldsymbol{J}(\boldsymbol{q}) = \partial \boldsymbol{f}(\boldsymbol{q})/\partial \boldsymbol{q}^T \in \mathbb{R}^{m \times n}$ represents the task Jacobian. It is assumed that the Jacobian $\boldsymbol{J}(\boldsymbol{q})$ has full row rank.

For non-redundant robots $(n = m)$, the operational space dynamics related to (48) can be described by

$$\boldsymbol{F} + \boldsymbol{F}_{\rm ext} = \boldsymbol{\Pi}(\boldsymbol{q})\ddot{\boldsymbol{p}} + \boldsymbol{\Gamma}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{p}} + \boldsymbol{\rho}(\boldsymbol{q}) + \boldsymbol{\zeta}(\boldsymbol{q}, \dot{\boldsymbol{q}}, t), \tag{49}$$

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where $\boldsymbol{F} \in \mathbb{R}^m$ denotes the commanded forces, and $\boldsymbol{F}_{\text{ext}} \in \mathbb{R}^m$ denotes the external forces; $\boldsymbol{\Pi} = \boldsymbol{J}^{-T}\boldsymbol{M}\boldsymbol{J}^{-1}$, $\boldsymbol{\Gamma} = \boldsymbol{J}^{-T}(\boldsymbol{C} - \boldsymbol{M}\boldsymbol{J}^{-1}\dot{\boldsymbol{J}})\boldsymbol{J}^{-1}$, and $\boldsymbol{\rho} = \boldsymbol{J}^{-T}\boldsymbol{g}$ denote the operational space inertia matrix, the Coriolis/centrifugal matrix, and the gravitational forces, respectively, and $\boldsymbol{\zeta} = \boldsymbol{J}^{-T}\boldsymbol{d}$ stands for the disturbance.

Importantly, the property (6) holds for the operational space dynamics (49), as follows:

$$\dot{\boldsymbol{\Pi}}(\boldsymbol{q}, \dot{\boldsymbol{q}}) = \boldsymbol{\Gamma}(\boldsymbol{q}, \dot{\boldsymbol{q}}) + \boldsymbol{\Gamma}^T(\boldsymbol{q}, \dot{\boldsymbol{q}}).$$

Hence, the control design and analyses presented in Sections III, IV, and V, and the Appendix carry over directly to the operational space formulation. As a result, the nonlinear $\mathcal{H}_\infty$ impedance control in operational space for non-redundant robots is derived as

$$\begin{aligned}
&\boldsymbol{F} = \widehat{\boldsymbol{\Pi}}\ddot{\boldsymbol{p}}_{\text{ref}} + \widehat{\boldsymbol{\Gamma}}\dot{\boldsymbol{p}}_{\text{ref}} + \widehat{\boldsymbol{\rho}} - \widehat{\boldsymbol{F}}_{\text{ext}} + \alpha\left(\boldsymbol{K}_\gamma + \frac{1}{\gamma^2}\boldsymbol{I}\right)\dot{\boldsymbol{\xi}}_{\text{ref}},\\
&\boldsymbol{\tau} = \boldsymbol{J}^T\boldsymbol{F},
\end{aligned} \tag{50}$$

where the dependence on $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ is omitted for brevity, and

$$\begin{aligned}
\ddot{\boldsymbol{p}}_{\text{ref}} &= \ddot{\boldsymbol{p}}_{\text{des}} + \boldsymbol{\Lambda}^{-1}\left(\boldsymbol{D}\dot{\boldsymbol{\xi}} + \boldsymbol{K}\boldsymbol{\xi} + \widehat{\boldsymbol{F}}_{\text{ext}}\right),\\
\dot{\boldsymbol{p}}_{\text{ref}} &= \dot{\boldsymbol{p}}_{\text{des}} + \boldsymbol{\Lambda}^{-1}\left(\boldsymbol{D}\boldsymbol{\xi} + \boldsymbol{K}\int\boldsymbol{\xi} + \int\widehat{\boldsymbol{F}}_{\text{ext}}\right),\\
\dot{\boldsymbol{\xi}}_{\text{ref}} &= \dot{\boldsymbol{\xi}} + \boldsymbol{\Lambda}^{-1}\left(\boldsymbol{D}\boldsymbol{\xi} + \boldsymbol{K}\int\boldsymbol{\xi} + \int\widehat{\boldsymbol{F}}_{\text{ext}}\right),
\end{aligned} \tag{51}$$

with the task error $\boldsymbol{\xi} = \boldsymbol{p}_{\text{des}} - \boldsymbol{p}$. Here, with a slight abuse of notation, the symbols α, γ, $\boldsymbol{K}_\gamma$, $\boldsymbol{\Lambda}$, $\boldsymbol{D}$, and $\boldsymbol{K}$ are reused to denote their operational space counterparts.

If the robot is kinematically redundant with respect to the given task, i.e., $n > m$, the kinematics (48) can be extended as follows:

$$\begin{bmatrix}\dot{\boldsymbol{p}}\\ \boldsymbol{z}\end{bmatrix} = \begin{bmatrix}\boldsymbol{J}(\boldsymbol{q})\\ \boldsymbol{Z}(\boldsymbol{q})\boldsymbol{W}(\boldsymbol{q})\end{bmatrix}\dot{\boldsymbol{q}} =: \overline{\boldsymbol{J}}(\boldsymbol{q})\dot{\boldsymbol{q}}, \tag{52}$$

where $\boldsymbol{Z}(\boldsymbol{q}) \in \mathbb{R}^{(n-m)\times n}$ denotes the null space basis[5] of the full-row-rank Jacobian $\boldsymbol{J}(\boldsymbol{q})$ such that $\boldsymbol{J}(\boldsymbol{q})\boldsymbol{Z}^T(\boldsymbol{q}) = \boldsymbol{0}$ holds, $\boldsymbol{z} \in \mathbb{R}^{n-m}$ denotes the corresponding null space velocities, and $\boldsymbol{W}(\boldsymbol{q}) \in \mathbb{R}^{n\times n}$ denotes the weight matrix. For a positive definite weight matrix $\boldsymbol{W}(\boldsymbol{q}) > \boldsymbol{0}$, e.g., $\boldsymbol{W} = \boldsymbol{I}$, the extended Jacobian $\overline{\boldsymbol{J}}(\boldsymbol{q}) \in \mathbb{R}^{n\times n}$ in (52) is always invertible [27]:

$$\overline{\boldsymbol{J}}^{-1} = \begin{bmatrix}\boldsymbol{J}^{W+} & \boldsymbol{Z}^{W\#}\end{bmatrix},$$

where $\boldsymbol{J}^{W+} := \boldsymbol{W}^{-1}\boldsymbol{J}^T(\boldsymbol{J}\boldsymbol{W}^{-1}\boldsymbol{J}^T)^{-1}$ denotes the weighted pseudo-inverse of $\boldsymbol{J}$, and $\boldsymbol{Z}^{W\#} := \boldsymbol{Z}^T(\boldsymbol{Z}\boldsymbol{W}\boldsymbol{Z}^T)^{-1}$.

The operational space dynamics related to the extended kinematics (52) can be described by

$$\begin{bmatrix}\boldsymbol{F}\\ \boldsymbol{\eta}\end{bmatrix} + \begin{bmatrix}\boldsymbol{F}_{\text{ext}}\\ \boldsymbol{\eta}_{\text{ext}}\end{bmatrix} = \overline{\boldsymbol{\Pi}}(\boldsymbol{q})\begin{bmatrix}\ddot{\boldsymbol{p}}\\ \dot{\boldsymbol{z}}\end{bmatrix} + \overline{\boldsymbol{\Gamma}}(\boldsymbol{q}, \dot{\boldsymbol{q}})\begin{bmatrix}\dot{\boldsymbol{p}}\\ \boldsymbol{z}\end{bmatrix} + \overline{\boldsymbol{\rho}}(\boldsymbol{q}) + \overline{\boldsymbol{\zeta}}(\boldsymbol{q}, \dot{\boldsymbol{q}}, t), \tag{53}$$

where $\boldsymbol{\eta} \in \mathbb{R}^{n-m}$ and $\boldsymbol{\eta}_{\text{ext}} \in \mathbb{R}^{n-m}$ denote the commanded and external generalized forces conjugate to the null space velocities $\boldsymbol{z}$, respectively; $\overline{\boldsymbol{\Pi}} = \overline{\boldsymbol{J}}^{-T}\boldsymbol{M}\overline{\boldsymbol{J}}^{-1}$, $\overline{\boldsymbol{\Gamma}} = \overline{\boldsymbol{J}}^{-T}(\boldsymbol{C} - \boldsymbol{M}\overline{\boldsymbol{J}}^{-1}\dot{\overline{\boldsymbol{J}}})\overline{\boldsymbol{J}}^{-1}$, $\overline{\boldsymbol{\rho}} = \overline{\boldsymbol{J}}^{-T}\boldsymbol{g}$, and $\overline{\boldsymbol{\zeta}} = \overline{\boldsymbol{J}}^{-T}\boldsymbol{d}$ satisfy

$$\begin{aligned}
\overline{\boldsymbol{\Pi}} &= \begin{bmatrix}\boldsymbol{\Pi}_p & \boldsymbol{\Pi}_{pz}\\ \boldsymbol{\Pi}_{pz}^T & \boldsymbol{\Pi}_z\end{bmatrix} = \begin{bmatrix}(\boldsymbol{J}^{W+})^T\boldsymbol{M}\boldsymbol{J}^{W+} & (\boldsymbol{J}^{W+})^T\boldsymbol{M}\boldsymbol{Z}^{W\#}\\ (\boldsymbol{Z}^{W\#})^T\boldsymbol{M}\boldsymbol{J}^{W+} & (\boldsymbol{Z}^{W\#})^T\boldsymbol{M}\boldsymbol{Z}^{W\#}\end{bmatrix},\\
\overline{\boldsymbol{\Gamma}} &= \begin{bmatrix}\boldsymbol{\Gamma}_p & \boldsymbol{\Gamma}_{pz}\\ \boldsymbol{\Gamma}_{zp} & \boldsymbol{\Gamma}_z\end{bmatrix} = \begin{bmatrix}(\boldsymbol{J}^{W+})^T\boldsymbol{C}_z\boldsymbol{J}^{W+} & (\boldsymbol{J}^{W+})^T\boldsymbol{C}_z\boldsymbol{Z}^{W\#}\\ (\boldsymbol{Z}^{W\#})^T\boldsymbol{C}_z\boldsymbol{J}^{W+} & (\boldsymbol{Z}^{W\#})^T\boldsymbol{C}_z\boldsymbol{Z}^{W\#}\end{bmatrix},\\
\overline{\boldsymbol{\rho}} &= \begin{bmatrix}\boldsymbol{\rho}_p\\ \boldsymbol{\rho}_z\end{bmatrix} = \begin{bmatrix}(\boldsymbol{J}^{W+})^T\boldsymbol{g}\\ (\boldsymbol{Z}^{W\#})^T\boldsymbol{g}\end{bmatrix}, \quad \text{and} \quad \overline{\boldsymbol{\zeta}} = \begin{bmatrix}\boldsymbol{\zeta}_p\\ \boldsymbol{\zeta}_z\end{bmatrix} = \begin{bmatrix}(\boldsymbol{J}^{W+})^T\boldsymbol{d}\\ (\boldsymbol{Z}^{W\#})^T\boldsymbol{d}\end{bmatrix},
\end{aligned}$$

in which $\boldsymbol{C}_z := \boldsymbol{C} - \boldsymbol{M}(\boldsymbol{J}^{W+}\dot{\boldsymbol{J}} + \boldsymbol{Z}^{W\#}(\dot{\boldsymbol{Z}}\boldsymbol{W} + \boldsymbol{Z}\dot{\boldsymbol{W}}))$.

The property (6) holds for the extended operational space dynamics (53) and, notably, for each of its diagonal blocks, i.e., each block also inherits this structural property of Euler–Lagrange systems:

$$\dot{\overline{\boldsymbol{\Pi}}} = \overline{\boldsymbol{\Gamma}} + \overline{\boldsymbol{\Gamma}}^T, \quad \dot{\boldsymbol{\Pi}}_p = \boldsymbol{\Gamma}_p + \boldsymbol{\Gamma}_p^T, \quad \dot{\boldsymbol{\Pi}}_z = \boldsymbol{\Gamma}_z + \boldsymbol{\Gamma}_z^T.$$

Hence, the nonlinear $\mathcal{H}_\infty$ impedance control in operational space for redundant robots is derived as

$$\begin{aligned}
&\boldsymbol{F} = \widehat{\boldsymbol{\Pi}}_p\ddot{\boldsymbol{p}}_{\text{ref}} + \widehat{\boldsymbol{\Gamma}}_p\dot{\boldsymbol{p}}_{\text{ref}} + \widehat{\boldsymbol{\rho}}_p - \widehat{\boldsymbol{F}}_{\text{ext}} + \alpha\left(\boldsymbol{K}_\gamma + \frac{1}{\gamma^2}\boldsymbol{I}\right)\dot{\boldsymbol{\xi}}_{\text{ref}},\\
&\boldsymbol{\tau} = \boldsymbol{J}^T\boldsymbol{F} + \boldsymbol{W}\boldsymbol{Z}^T\boldsymbol{\eta},
\end{aligned}$$

where the dependence on $\boldsymbol{q}$ and $\dot{\boldsymbol{q}}$ is omitted for brevity, and $\ddot{\boldsymbol{p}}_{\text{ref}}$, $\dot{\boldsymbol{p}}_{\text{ref}}$, and $\dot{\boldsymbol{\xi}}_{\text{ref}}$ are given by (51). It is noted that, although the null dynamics also takes the form of an Euler–Lagrange system, imposing a desired impedance directly on the null space via RMC with REF is somewhat ambiguous due to the nonholonomic nature of the null space velocities [28]. One remedy is to define the desired impedance model in a secondary task space and then map it into the null space of the primary task [29], [30]. It is also noted that, when $\boldsymbol{W} = \boldsymbol{M}$,[6] the well-known property of dynamic consistency ensures that the null forces $\boldsymbol{\eta}$ do not affect the task acceleration [29], [30]. However, in practical implementations, $\boldsymbol{W} = \widehat{\boldsymbol{M}}$ is typically employed, and therefore perfect inertia decoupling cannot be achieved in the presence of parametric uncertainties [29]. Furthermore, the Coriolis coupling generally remains. Fortunately, the residual coupling between the task and null dynamics can be treated as an additional disturbance and robustly attenuated by the REF.

VII. Experimental Results

To demonstrate the effectiveness of the proposed $\mathcal{H}_\infty$ impedance control (50), several experiments were conducted using a 6-DOF torque-controlled robot (Indy7, Neurometka) equipped with a 6-DOF force/torque (F/T) sensor (AFT200-D80-C, Aidin Robotics), as shown in Fig. 4. The robot features high gear reduction ratios (121:1 for joints 1–3 and 101:1 for joints 4–6), and therefore it experiences substantial joint friction, which makes physical interaction more challenging. The robot has a rated payload of 7 kg. The weights of the F/T sensor and tools were not modeled and therefore acted as disturbances. In all experiments, the control input torques

[5]The null space basis matrix can be obtained, for instance, via singular value decomposition. In this case, $\boldsymbol{Z}\boldsymbol{Z}^T = \boldsymbol{I}$ holds, and the time derivative of $\boldsymbol{Z}$ can be chosen as $\dot{\boldsymbol{Z}} = -\boldsymbol{Z}\dot{\boldsymbol{J}}^T(\boldsymbol{J}^+)^T$, where $(\cdot)^+$ denotes the Moore–Penrose pseudo-inverse of $(\cdot)$.

[6]In fact, there exist infinitely many configuration-dependent weight matrices $\boldsymbol{W}(\boldsymbol{q})$ that ensure dynamic consistency [29], all of which involve the joint space inertia matrix $\boldsymbol{M}(\boldsymbol{q})$.

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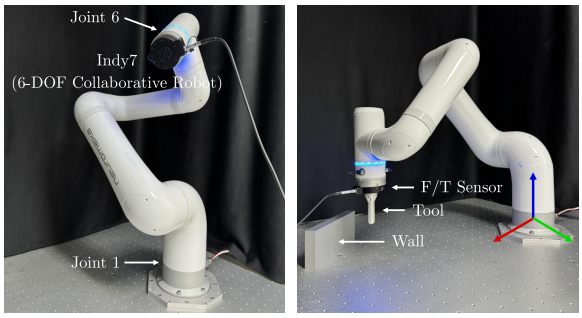


Fig. 4. (Left) 6-DOF collaborative robot used in all experiments. (Right) Experimental setup for Experiment B.

were computed at a control frequency of 4 kHz, while the force signals measured by the F/T sensor were updated at a sampling rate of 1 kHz, with no filtering applied.

In all experiments, the control was performed in operational space, where the task coordinates are defined as

$$\boldsymbol{p} = \begin{bmatrix} x & y & z & \theta_x & \theta_y & \theta_z \end{bmatrix}^T \in \mathbb{R}^6$$

encompassing both position and orientation of the end-effector, with the orientation represented by Euler ZYX angles, and the corresponding task forces are defined as

$$\boldsymbol{F} = \begin{bmatrix} f_x & f_y & f_z & m_x & m_y & m_z \end{bmatrix}^T \in \mathbb{R}^6.$$

For fair comparisons, all controllers were configured with identical impedance (admittance) parameters in each experiment. Throughout all experiments, the proposed controller (50) was implemented with $\alpha = 1$ (except for Experiment F), while the $\mathcal{L}_2$-gain γ and the gain matrix $\boldsymbol{K}_\gamma$ were set as

$$\gamma = 0.1, \quad \boldsymbol{K}_\gamma = \mathrm{diag}(1000, 1000, 1000, 50, 50, 50).$$

Note that these gains are specified in operational space.

The performance of the proposed controller was compared with that of conventional impedance and admittance control. In addition, two advanced compliant controllers were included for comparison: admittance control with bidirectional energy flow modulation [21] (passive admittance control), and sliding mode impedance control [8] (robust impedance control). A brief description of each controller is provided below.

1) Impedance control: The computed-torque-based implementation was adopted for impedance control, since the proposed $\mathcal{H}_\infty$ impedance control also employs a modified computed torque control (RMC) with an additional compensation term (REF). The impedance control is given by

$$\begin{aligned}\boldsymbol{\tau} = \widehat{\boldsymbol{M}}(\boldsymbol{q})\boldsymbol{J}^{-1}(\boldsymbol{q})(\ddot{\boldsymbol{p}}_{\mathrm{ref}} - \dot{\boldsymbol{J}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}}) + \widehat{\boldsymbol{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \widehat{\boldsymbol{g}}(\boldsymbol{q}) \\ - \boldsymbol{J}^T(\boldsymbol{q})\widehat{\boldsymbol{F}}_{\mathrm{ext}},\end{aligned} \tag{54}$$

where the reference acceleration is given by

$$\ddot{\boldsymbol{p}}_{\mathrm{ref}} = \ddot{\boldsymbol{p}}_{\mathrm{des}} + \boldsymbol{\Lambda}^{-1}(\boldsymbol{D}\dot{\boldsymbol{\xi}} + \boldsymbol{K}\boldsymbol{\xi} + \widehat{\boldsymbol{F}}_{\mathrm{ext}}). \tag{55}$$

2) Admittance control: Admittance control was implemented based on the NRIC framework, since it can be viewed as a nonlinear $\mathcal{H}_\infty$ impedance control, as discussed in Section I. In classical admittance control, the nominal model is typically defined as a single lumped body in task space [26], [31]. This requires the inner-loop position controller to be designed in task space, and such a controller can be nonlinear and complex for multi-DOF robots. Alternatively, in the NRIC-based admittance control, the nominal model is defined as a virtual articulated multibody robot whose kinematic parameters are identical to those of the real robot. Since the nominal robot can generate the reference joint trajectory, a simple joint space tracking controller can be used as the position controller.

The nominal robot dynamics is described by

$$\boldsymbol{\tau}_n + \boldsymbol{J}^T(\boldsymbol{q}_n)\widehat{\boldsymbol{F}}_{\mathrm{ext}} = \boldsymbol{M}_n(\boldsymbol{q}_n)\ddot{\boldsymbol{q}}_n + \boldsymbol{C}_n(\boldsymbol{q}_n,\dot{\boldsymbol{q}}_n)\dot{\boldsymbol{q}}_n + \boldsymbol{g}_n(\boldsymbol{q}_n),$$

where $\boldsymbol{q}_n$ and $\dot{\boldsymbol{q}}_n$ are the nominal robot's states, both in $\mathbb{R}^n$. Here, there are no restrictions on the design of the nominal robot's dynamic parameters $\boldsymbol{M}_n$, $\boldsymbol{C}_n$, and $\boldsymbol{g}_n$. Nevertheless, in our implementation, they were chosen as $\boldsymbol{M}_n(\boldsymbol{q}_n) = \widehat{\boldsymbol{M}}(\boldsymbol{q}_n)$, $\boldsymbol{C}_n(\boldsymbol{q}_n,\dot{\boldsymbol{q}}_n) = \widehat{\boldsymbol{C}}(\boldsymbol{q}_n,\dot{\boldsymbol{q}}_n)$, and $\boldsymbol{g}_n(\boldsymbol{q}_n) = \widehat{\boldsymbol{g}}(\boldsymbol{q}_n)$. Note that $\widehat{\boldsymbol{M}}(\boldsymbol{q}_n)$, $\widehat{\boldsymbol{C}}(\boldsymbol{q}_n,\dot{\boldsymbol{q}}_n)$, and $\widehat{\boldsymbol{g}}(\boldsymbol{q}_n)$ differ from the estimated robot dynamics $\widehat{\boldsymbol{M}}(\boldsymbol{q})$, $\widehat{\boldsymbol{C}}(\boldsymbol{q},\dot{\boldsymbol{q}})$, and $\widehat{\boldsymbol{g}}(\boldsymbol{q})$, since they are dependent on the nominal robot's states.

Impedance control for the nominal robot is designed as

$$\begin{aligned}\boldsymbol{\tau}_n = \widehat{\boldsymbol{M}}(\boldsymbol{q}_n)\boldsymbol{J}^{-1}(\boldsymbol{q}_n)(\ddot{\boldsymbol{p}}_n^{\mathrm{ref}} - \dot{\boldsymbol{J}}(\boldsymbol{q}_n,\dot{\boldsymbol{q}}_n)\dot{\boldsymbol{q}}_n) + \widehat{\boldsymbol{C}}(\boldsymbol{q}_n,\dot{\boldsymbol{q}}_n)\dot{\boldsymbol{q}}_n \\ + \widehat{\boldsymbol{g}}(\boldsymbol{q}_n) - \boldsymbol{J}^T(\boldsymbol{q}_n)\widehat{\boldsymbol{F}}_{\mathrm{ext}},\end{aligned}$$

where $\ddot{\boldsymbol{p}}_n^{\mathrm{ref}} = \ddot{\boldsymbol{p}}_{\mathrm{des}} + \boldsymbol{\Lambda}^{-1}(\boldsymbol{D}\dot{\boldsymbol{\xi}}_{dn} + \boldsymbol{K}\boldsymbol{\xi}_{dn} + \widehat{\boldsymbol{F}}_{\mathrm{ext}})$ with the nominal tracking error $\boldsymbol{\xi}_{dn} = \boldsymbol{p}_{\mathrm{des}} - \boldsymbol{p}_n$, in which $\boldsymbol{p}_n = \boldsymbol{f}(\boldsymbol{q}_n)$. Since the nominal robot is free from modeling uncertainties, it can render the desired impedance behavior as

$$\boldsymbol{\Lambda}\ddot{\boldsymbol{\xi}}_{dn} + \boldsymbol{D}\dot{\boldsymbol{\xi}}_{dn} + \boldsymbol{K}\boldsymbol{\xi}_{dn} = -\widehat{\boldsymbol{F}}_{\mathrm{ext}}.$$

Using the nominal control input $\boldsymbol{\tau}_n$, the nominal robot is simulated in real time via forward dynamics $\ddot{\boldsymbol{q}}_n = \boldsymbol{M}_n^{-1}(\boldsymbol{\tau}_n + \boldsymbol{J}^T\widehat{\boldsymbol{F}}_{\mathrm{ext}} - \boldsymbol{C}_n\dot{\boldsymbol{q}}_n - \boldsymbol{g}_n)$ and numerical integration. In our implementation, the fourth-order Runge–Kutta method was employed for the numerical integration. Subsequently, the following PID control input $\boldsymbol{\tau}_a$ forces the real robot to closely track the simulated nominal robot:

$$\boldsymbol{\tau}_a = -\left(\boldsymbol{K}_\gamma' + \frac{1}{\gamma'^2}\boldsymbol{I}\right)\left(\dot{\boldsymbol{e}}_{nr} + \boldsymbol{K}_p'\boldsymbol{e}_{nr} + \boldsymbol{K}_i'\int \boldsymbol{e}_{nr}\right), \tag{56}$$

where $\boldsymbol{e}_{nr} = \boldsymbol{q}_n - \boldsymbol{q}$. Throughout all experiments, the gains were configured as

$$\begin{aligned}\boldsymbol{K}_\gamma' + (1/\gamma'^2)\boldsymbol{I} &= \mathrm{diag}(1200, 1200, 600, 400, 400, 400), \\ \boldsymbol{K}_p' &= \mathrm{diag}(60, 60, 40, 30, 30, 30), \\ \boldsymbol{K}_i' &= \mathrm{diag}(5, 5, 5, 5, 5, 5).\end{aligned}$$

Note that these gains are specified in joint space.

Within the NRIC framework, the nominal control input $\boldsymbol{\tau}_n$ is applied to the real robot as a feedforward input, as shown in Fig. 1. For linear systems, the proper use of a feedforward term can improve the position control bandwidth and stability margin [26], [32]. In our implementation, the inner-loop bandwidths for joints 1–6 are approximately 17, 13, 23, 63, 69, and 69 Hz, respectively, at -3 dB. Finally, the NRIC-based admittance control is given by

$$\boldsymbol{\tau} = \boldsymbol{\tau}_n - \boldsymbol{\tau}_a. \tag{57}$$

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3) Admittance control with bidirectional energy flow modulation: In [21], an additional feedback connection, referred to as bidirectional energy flow, was introduced into the admittance control structure to suppress internally generated energy during physical interaction. This control scheme is also essentially based on the NRIC framework, where the nominal robot and the nominal controller are given as follows:

$$\boldsymbol{\tau}_n + \boldsymbol{\tau}_{bd} + \boldsymbol{J}^T(\boldsymbol{q}_n)\widehat{\boldsymbol{F}}_{\text{ext}} = \boldsymbol{M}_n(\boldsymbol{q}_n)\ddot{\boldsymbol{q}}_n + \boldsymbol{C}_n(\boldsymbol{q}_n, \dot{\boldsymbol{q}}_n)\dot{\boldsymbol{q}}_n,$$
$$\boldsymbol{\tau}_n = \boldsymbol{J}^T(\boldsymbol{q}_n)(\boldsymbol{K}\boldsymbol{\xi}_{dn} - \boldsymbol{D}\dot{\boldsymbol{p}}_n),$$

in which $\boldsymbol{M}_n(\boldsymbol{q}_n) = \lambda\widehat{\boldsymbol{M}}(\boldsymbol{q}_n)$, $\boldsymbol{C}_n(\boldsymbol{q}_n, \dot{\boldsymbol{q}}_n) = \lambda\widehat{\boldsymbol{C}}(\boldsymbol{q}_n, \dot{\boldsymbol{q}}_n)$, and $\boldsymbol{g}_n(\boldsymbol{q}_n) = \boldsymbol{0}$, with an inertia scaling factor $\lambda > 0$. Here, a PD-based impedance controller is adopted as the nominal controller. Unlike in the standard NRIC framework, the nominal robot is affected by the bidirectional energy flow torque $\boldsymbol{\tau}_{bd}$. The bidirectional energy flow torque $\boldsymbol{\tau}_{bd}$ and the control input torque $\boldsymbol{\tau}$ are given by

$$\boldsymbol{\tau}_{bd} = \lambda\beta\boldsymbol{\tau}_a, \qquad \boldsymbol{\tau} = \boldsymbol{\tau}_r - (1-\beta)\boldsymbol{\tau}_a, \tag{58}$$

where $\boldsymbol{\tau}_a$ is given by (56), and $\boldsymbol{\tau}_r$ is given by

$$\boldsymbol{\tau}_r = \widehat{\boldsymbol{g}}(\boldsymbol{q}) + (\lambda^{-1} - 1)\boldsymbol{J}^T(\boldsymbol{q})\widehat{\boldsymbol{F}}_{\text{ext}} + \lambda^{-1}\boldsymbol{J}^T(\boldsymbol{q})(\boldsymbol{K}\boldsymbol{\xi} - \boldsymbol{D}\dot{\boldsymbol{p}}).$$

When $\lambda = 1$, the feedforward input $\boldsymbol{\tau}_r$ reduces to a simple PD-based impedance control. In all experiments, $\lambda = 1$ was used, as in [21], since inertia reduction in admittance control can easily destabilize the system.

The controller operates as an admittance controller (with the feedforward input) when $\beta = 0$, and as a pure impedance controller when $\beta = 1$. Accordingly, a β-modulation rule ($0 \leq \beta \leq 1$) was proposed based on a passivity analysis to modulate the compensation torque $(1-\beta)\boldsymbol{\tau}_a$ and the bidirectional energy flow torque $\lambda\beta\boldsymbol{\tau}_a$. The main philosophy of the β-modulation rule is to increase β only to the minimum extent required to prevent energy generation, thereby minimizing the sacrifice of robustness inherent to admittance control. Details of the β-modulation rule can be found in [21].

4) Sliding mode impedance control: In [8], the integral sliding mode [33] and the second-order super-twisting algorithm [34] were integrated with impedance control to realize a robust impedance controller. The control law is given by

$$\boldsymbol{\tau} = \widehat{\boldsymbol{M}}(\boldsymbol{q})(\boldsymbol{v}_0 + \boldsymbol{v}_{\text{SMC}}) + \widehat{\boldsymbol{C}}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \widehat{\boldsymbol{g}}(\boldsymbol{q}) - \boldsymbol{J}^T(\boldsymbol{q})\widehat{\boldsymbol{F}}_{\text{ext}}, \tag{59}$$

where

$$\boldsymbol{v}_0 = \boldsymbol{J}^{-1}(\boldsymbol{q})(\ddot{\boldsymbol{p}}_{\text{ref}} - \dot{\boldsymbol{J}}(\boldsymbol{q}, \dot{\boldsymbol{q}})\dot{\boldsymbol{q}}),$$
$$\boldsymbol{v}_{\text{SMC}} = \boldsymbol{K}_1\sqrt{||\boldsymbol{\sigma}_q||}\,\text{sgn}(\boldsymbol{\sigma}_q) + \boldsymbol{K}_2\int \text{sgn}(\boldsymbol{\sigma}_q). \tag{60}$$

Here, the reference acceleration $\ddot{\boldsymbol{p}}_{\text{ref}}$ is given in (55), and the signum function and the sliding vector are defined by

$$\text{sgn}(\boldsymbol{\sigma}_q) = \frac{\boldsymbol{\sigma}_q}{||\boldsymbol{\sigma}_q|| + \varepsilon}, \quad \boldsymbol{\sigma}_q = \boldsymbol{J}^T(\boldsymbol{q})\boldsymbol{\sigma},$$
$$\boldsymbol{\sigma} = \boldsymbol{\Lambda}\dot{\boldsymbol{\xi}} + \boldsymbol{D}\boldsymbol{\xi} + \boldsymbol{K}\int \boldsymbol{\xi} + \int \widehat{\boldsymbol{F}}_{\text{ext}}, \tag{61}$$

where the boundary layer thickness ε is chosen to balance chattering reduction and sliding accuracy. In our implementation, $\varepsilon = 0.001$ was used, as in [8].

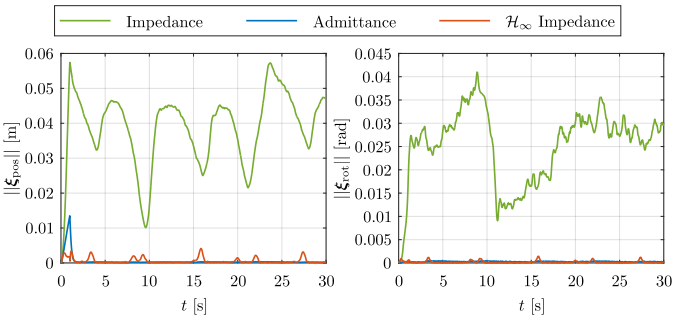


Fig. 5. Experimental results of Experiment A, where $(\cdot)_{\text{pos}}$ and $(\cdot)_{\text{rot}}$ denote the linear and angular components of $(\cdot)$, respectively.

The sliding vector $\boldsymbol{\sigma}$ (61) defines an integral sliding mode. The philosophy of the integral sliding mode is to eliminate the reaching phase by ensuring that the system remains on the sliding surface $\boldsymbol{\sigma} = \boldsymbol{0}$ from the initial time instant.

A. Free motion

The first experiment (Experiment A) compares the tracking performance of the proposed $\mathcal{H}_\infty$ impedance control with those of the impedance control (54) and the admittance control (57) in free space, where no physical interaction occurs, i.e., $\boldsymbol{F}_{\text{ext}} = \boldsymbol{0}$. In particular, to highlight robustness under low impedance gains, the impedance parameters were set as

$$\boldsymbol{\Lambda} = \text{diag}(1, 1, 1, 0.1, 0.1, 0.1),$$
$$\boldsymbol{D} = \text{diag}(20, 20, 20, 4, 4, 4),$$
$$\boldsymbol{K} = \text{diag}(100, 100, 100, 40, 40, 40).$$

The desired trajectory was specified such that the robot followed an infinity-shaped path for 30 s. The experimental results are shown in Fig. 5.

As expected, the admittance control exhibits excellent tracking performance during free motion. In contrast, the impedance control suffers from significant performance degradation due to joint friction and parametric uncertainties. Notably, despite the use of low impedance gains, the proposed $\mathcal{H}_\infty$ impedance control achieves satisfactory tracking performance, comparable to that of the admittance control.

It is worth noting that, in free space ($\boldsymbol{F}_{\text{ext}} = \boldsymbol{0}$), the proposed $\mathcal{H}_\infty$ impedance control via RMC with REF (39) reduces to the $\mathcal{H}_\infty$ position control via RMC with REF (15). For notational simplicity, the measured external force $\widehat{\boldsymbol{F}}_{\text{ext}}$ will be denoted by $\boldsymbol{F}_{\text{ext}}$ hereafter.

B. Physical interaction with stiff environment

In the following experiment (Experiment B), we compare the performance of the controllers during physical interaction with a stiff environment. To this end, a stiff wall was installed within the robot's workspace, and a rigid tool was mounted on the robot's end-effector, as illustrated in Fig. 4.

As in the previous experiment, the robot was commanded to follow an infinity-shaped trajectory; however, the wall was placed so as to obstruct the desired path. Consequently, the robot physically interacted with the wall through the tool

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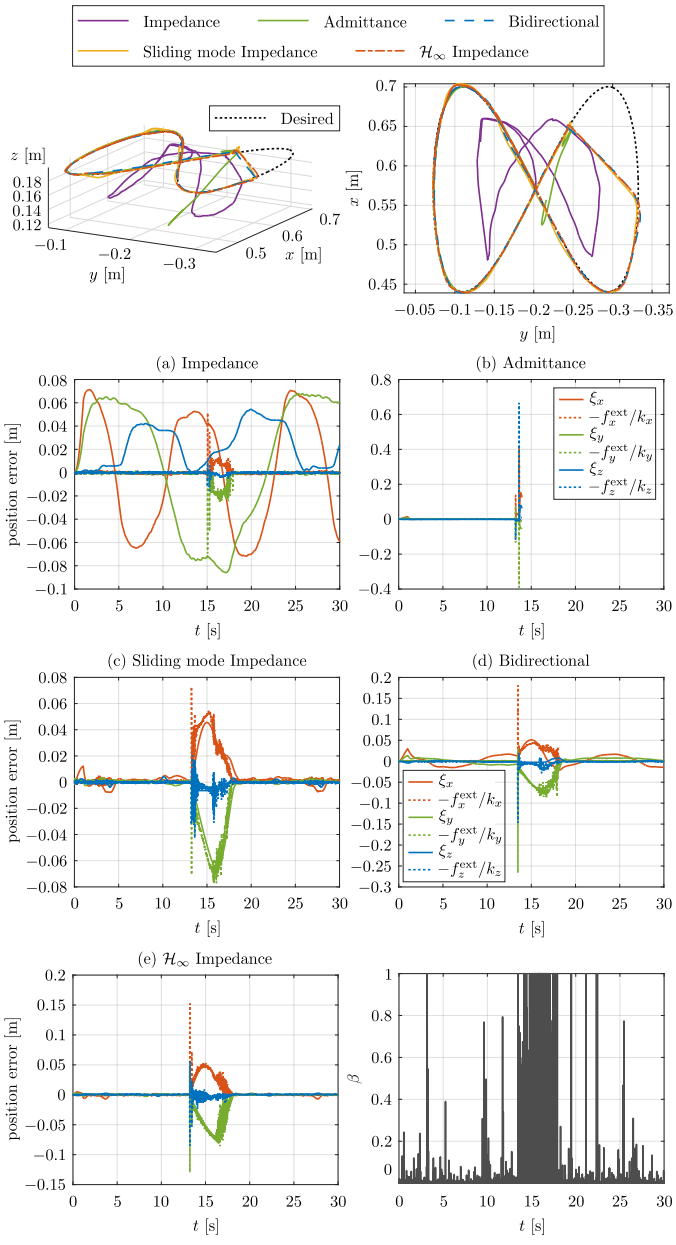


Fig. 6. Experimental results of Experiment B, where $(\cdot)_x$, $(\cdot)_y$, and $(\cdot)_z$ denote the x-, y-, and z-components of $(\cdot)$, respectively. For instance, $\boldsymbol{K} = \mathrm{diag}(k_x, k_y, k_z, k_{\theta_x}, k_{\theta_y}, k_{\theta_z})$.

and was required to adapt its motion to comply with the wall according to the specified impedance parameters. The impedance parameters were configured as follows:

$$\boldsymbol{\Lambda} = \mathrm{diag}(10, 10, 10, 1, 1, 1),$$
$$\boldsymbol{D} = \mathrm{diag}(80, 80, 80, 10, 10, 10),$$
$$\boldsymbol{K} = \mathrm{diag}(400, 400, 400, 50, 50, 50).$$

In this experiment, the admittance control with bidirectional energy flow modulation (58) and the sliding mode impedance control (59) were also included for comparison. The experimental results are shown in Fig. 6, where physical interaction with the wall occurs during $t \approx 13$–18 s.

First, the admittance control exhibits excellent tracking performance in free space, as observed in the previous experiment. However, upon physical contact with the wall, it generates excessively large contact forces and induces instability, resulting in an emergency stop for safety. It is well known that admittance control can easily lead to instability during physical interaction with stiff environments [1]–[3], [26].

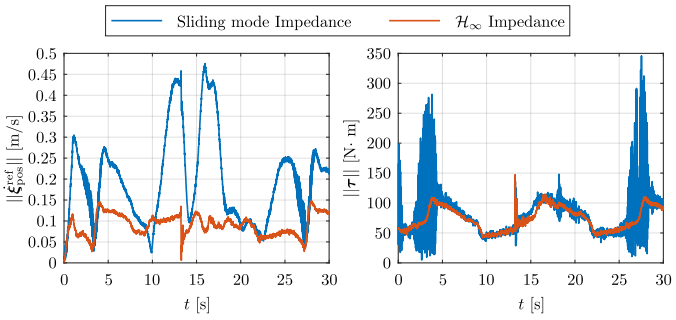


Fig. 7. (Continued) Experimental results of Experiment B.

The impedance control fails to overcome disturbances and thus exhibits poor tracking performance during free motion, similar to that observed in the previous experiment. During physical interaction with the wall, although it does not induce instability owing to the inherent impedance causality, it exhibits significant deviation from the equilibrium position $\boldsymbol{\xi} = -\boldsymbol{K}^{-1}\boldsymbol{F}_{\text{ext}}$ due to the lack of robustness.

In contrast, the proposed controller achieves excellent performance not only during free motion but also during physical interaction with the wall. In particular, it adapts to the wall in a stable manner while closely tracking the equilibrium position.

Next, we examine the results of the admittance control with bidirectional energy flow modulation. During the free motion phase, the modulation parameter β remains close to zero. As a result, the controller behaves similarly to admittance control and achieves good tracking performance during free motion. The slight performance degradation observed in free space is attributed to the fact that β does not remain exactly zero and that the control scheme is inherently based on the PD-based impedance control, which is designed for set-point regulation ($\ddot{\boldsymbol{p}}_{\text{des}} = \dot{\boldsymbol{p}}_{\text{des}} = \boldsymbol{0}$). During the adaptation phase, the parameter β approaches and remains close to one, causing the controller to behave similarly to the pure PD-based impedance control. Hence, stable interaction with the wall is achieved.

Finally, we examine the results of the sliding mode impedance control. In this experiment, the sliding mode control gains in (60) were set as

$$\boldsymbol{K}_1 = \mathrm{diag}(50, 50, 40, 40, 40, 40), \quad \boldsymbol{K}_2 = 0.1\boldsymbol{I}. \tag{62}$$

The sliding mode impedance control achieves good performance during both free motion and physical interaction, comparable to that of the proposed controller and the admittance control with bidirectional energy flow modulation.

We also investigate the sliding mode enforcement performance. As discussed in Sections III and IV, the proposed controller operates in a similar manner to SMC, and the sliding variable $\boldsymbol{s}$ in task space is given by $\boldsymbol{s} = \dot{\boldsymbol{\xi}}_{\text{ref}}$, as can be inferred from (37). Fortunately, both the proposed controller and the sliding mode impedance controller define the same sliding surface $\boldsymbol{s} = \boldsymbol{0}$, since $\boldsymbol{\sigma} = \boldsymbol{\Lambda}\boldsymbol{s}$. Therefore, their sliding mode

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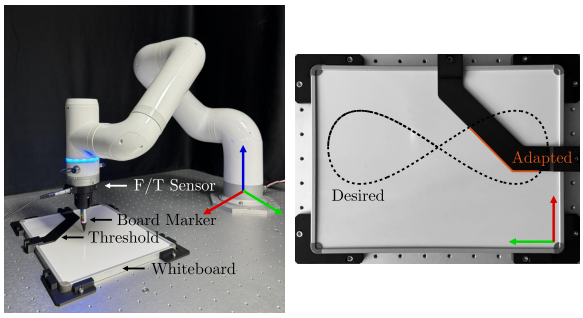


Fig. 8. Experimental setup for Experiment C.

performance can be directly compared. The corresponding results are presented in Fig. 7.

The sliding mode impedance controller exhibits degraded sliding mode performance compared with the proposed controller over the entire experimental interval. This result is attributed to the relatively low sliding mode control gains used in this experiment. However, increasing the gains was not feasible in practice because higher gains induced severe torque chattering and repeatedly led to loss of contact and bouncing off the wall, despite the high control frequency of 4 kHz used in our implementation. Even with the reduced gains, noticeable chattering in the control torque is still observed in certain configurations, as shown in Fig. 7.

C. Hybrid impedance control

In this experiment (Experiment C), we compare the performance in a hybrid control scenario, which is representative of robotic tasks such as polishing. In such tasks, the constrained direction is controlled in force, while the unconstrained direction is controlled in motion. The desired dynamics in the hybrid impedance control is expressed as follows:

$$\boldsymbol{\Lambda}\ddot{\boldsymbol{\xi}} + \boldsymbol{D}\dot{\boldsymbol{\xi}} + \boldsymbol{K}\boldsymbol{\xi} = \boldsymbol{F}_{\mathrm{des}} - \boldsymbol{F}_{\mathrm{ext}}.$$

The experiment was designed such that the robot draws an infinity symbol (∞) on a whiteboard using a board marker attached to its end-effector, as illustrated in Fig. 8. Pure force control was applied along the z-axis (normal to the whiteboard surface) with a desired force profile $f_z^{\mathrm{des}}(t) = 10 + 3\sin(t)$ N, while motion control was performed in the other directions to follow the desired path. To induce further physical interaction along the motion directions, a threshold was placed to obstruct the desired path, requiring the robot to adapt its motion along the threshold. The impedance parameters were specified as

$$\begin{aligned}\boldsymbol{\Lambda} &= \mathrm{diag}(4, 4, 10, 0.4, 0.4, 0.4),\\ \boldsymbol{D} &= \mathrm{diag}(80, 80, 200, 80, 80, 80),\\ \boldsymbol{K} &= \mathrm{diag}(400, 400, 0, 400, 400, 400),\end{aligned}$$

where zero stiffness was set along the z-axis for force control. In this experiment, the admittance control with and without bidirectional energy flow modulation, and the proposed controller were evaluated. The experimental results are presented in Fig. 9, where the robot adapts its motion along the threshold during $t \approx 13$–19 s.

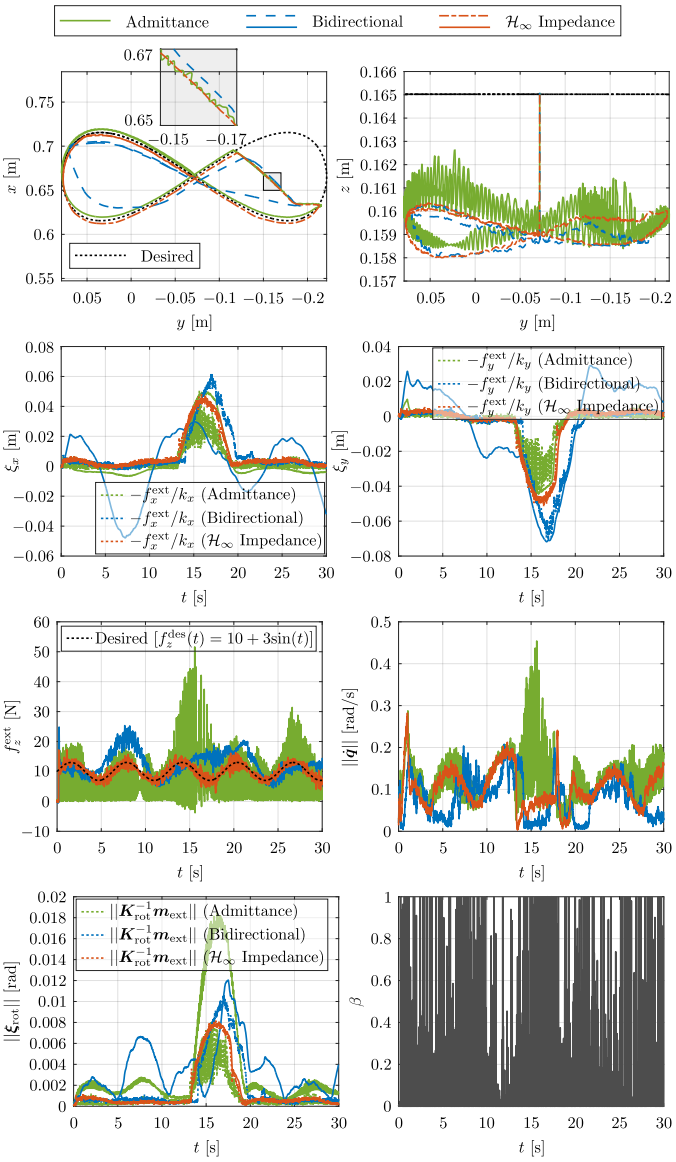


Fig. 9. Experimental results of Experiment C.

From the beginning of the task, the admittance control exhibits persistent chattering in the force-controlled direction despite the relatively high damping assigned along the z-axis. As a result, the admittance control fails to track the desired force profile throughout the entire experiment. Additionally, during the adaptation phase, the admittance control fails to achieve stable adaptation to the threshold and exhibits chattering due to the low inertia and damping assigned along the x- and y-axes, despite the relatively soft tip of the board marker. It is well known that admittance control is prone to oscillation and instability under low inertia and damping [26], [35].

The proposed controller and the admittance control with bidirectional energy flow modulation maintain stable contact with the whiteboard surface and the threshold. However, in the admittance control with bidirectional energy flow modulation, the parameter β deviates considerably from zero and, in particular, remains close to one during the adaptation phase, which leads to significant performance degradation. In contrast, the

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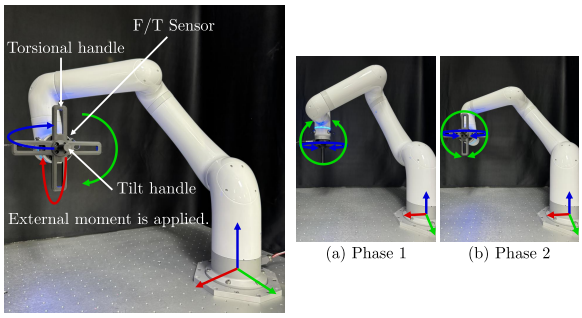


Fig. 10. Experimental setup for Experiment D. In phase 1, the torsional handle motion (the rotation of joint 6, i.e., q_6) is mainly affected by the z-axis stiffness, whereas in phase 2, it is primarily governed by the y-axis stiffness.

proposed controller consistently renders the desired impedance in both force- and motion-controlled directions, demonstrating accurate force tracking with stable threshold adaptation.

It is worth mentioning that the pronounced chattering observed with the admittance control can be mitigated by increasing the damping. Nevertheless, a low-damping setting was deliberately chosen to highlight that the proposed controller remains stable even under low damping, owing to the highly dissipative nature of the REF, as discussed in Section V.

D. Rotational impedance rendering

This experiment (Experiment D) evaluates the performance of Cartesian rotational impedance rendering. The experimental setup is illustrated in Fig. 10. The robot was commanded to maintain its initial configuration (set-point regulation), and a human operator grasped the handle and applied external moments to the robot. The impedance parameters were set as

$$\begin{aligned}\boldsymbol{\Lambda} &= \mathrm{diag}(4, 4, 4, 2, 0.4, 0.4),\\ \boldsymbol{D} &= \mathrm{diag}(200, 200, 200, 20, 2, 4),\\ \boldsymbol{K} &= \mathrm{diag}(2000, 2000, 2000, 0, 200, 10),\end{aligned}$$

where the translational impedance was set to high values to prevent linear motion, and the rotational impedance was designed differently for each rotational axis: zero stiffness for roll (θ_x), high stiffness with underdamping for pitch (θ_y), and low stiffness with critical damping for yaw (θ_z). The experimental scenario consists of two phases, as illustrated in Fig. 10: In phase 1, the human operator applies angular moments along the y- and z-axes, and the resulting behaviors are observed. Subsequently, the operator applies a moment along the x-axis until the roll angle reaches approximately 90 deg. Due to the zero stiffness along the x-axis, the robot does not restore its initial configuration and instead fully complies with the operator (phase 1 $\rightarrow$ phase 2). In phase 2, the operator again applies angular moments along the y- and z-axes, and the resulting behaviors are examined. Through this scenario, it can be verified whether Cartesian rotational impedance, rather than joint space impedance, is properly rendered.

In this experiment, the performance of four controllers—impedance, admittance, sliding mode impedance, and $\mathcal{H}_\infty$ impedance—was compared, where the sliding mode control gains were set as in (62). Since this experiment involves human–robot interaction (HRI), ten participants performed three trials for each controller (30 trials per controller) to mitigate human bias and trial-to-trial variability, and the performance was compared statistically. To ensure fairness, the participants were not informed of which controller was being used. The experimental results are shown in Figs. 11 and 12.

We first examine the impedance control. Along the z-axis, when external moments are applied and subsequently released, the impedance control fails to overcome joint friction due to the low stiffness, so that the robot does not return to its initial configuration and instead converges to a different equilibrium. Along the y-axis, despite the high stiffness, the impedance control responds in a compliant manner owing to the lack of robustness, resulting in a substantial deviation from the equilibrium position $\xi_{\theta_y} = -m_y^{\mathrm{ext}}/k_{\theta_y}$.

Next, we examine the results of the admittance control. Unlike the impedance control, when external moments along the y- and z-axes are applied and subsequently released, the admittance control successfully restores the robot to its initial configuration, owing to its inherent robustness. However, the actual robot response $\boldsymbol{\xi}_{\mathrm{rot}}$ significantly overshoots the intended equilibrium $-\boldsymbol{K}_{\mathrm{rot}}^{-1}\boldsymbol{m}_{\mathrm{ext}}$. The participants consistently reported that the robot felt "lightweight" when controlled by the admittance control. This behavior can be attributed to the causality of admittance control: external forces are mapped to position commands via numerical integration of the nominal model, which can produce exaggerated position commands due to its inertial and damping effects [19]. In several trials, the admittance control induced instability during physical interaction. Such instability of admittance control in HRI, particularly when human operators employ a firm grasp or stiffen their arms, has been widely reported [26], [31], [35].

We then examine the results of the proposed controller. Throughout the entire experiment, the proposed controller enables the robot to comply smoothly with the human in accordance with the specified impedance parameters, following the equilibrium position. Upon release of the external moments along the y- and z-axes, the robot restores its initial configuration while rendering underdamped behavior along the y-axis and critically damped behavior along the z-axis.

We now examine the sliding mode impedance control. When external moments along the y- and z-axes are applied and subsequently released, the sliding mode impedance control robustly achieves zero steady-state error. However, along the z-axis, transient overshoot is observed despite the critical-damping setting. This can be attributed to the super-twisting algorithm (60) with the integral sliding surface (61). In particular, it has been reported that second-order super-twisting algorithms based on integral sliding surfaces, while capable of fast response, may exhibit transient overshoot [36].

From the statistical analysis of the 30 trials shown in Fig. 12, along the z-axis with low stiffness and critical damping, the sliding mode impedance control exhibits performance comparable to that of the proposed controller. This result suggests that the sliding mode impedance control responds more rapidly to externally applied moments than the proposed controller, albeit at the expense of transient overshoot. Along the y-axis with high stiffness and underdamping, the proposed controller

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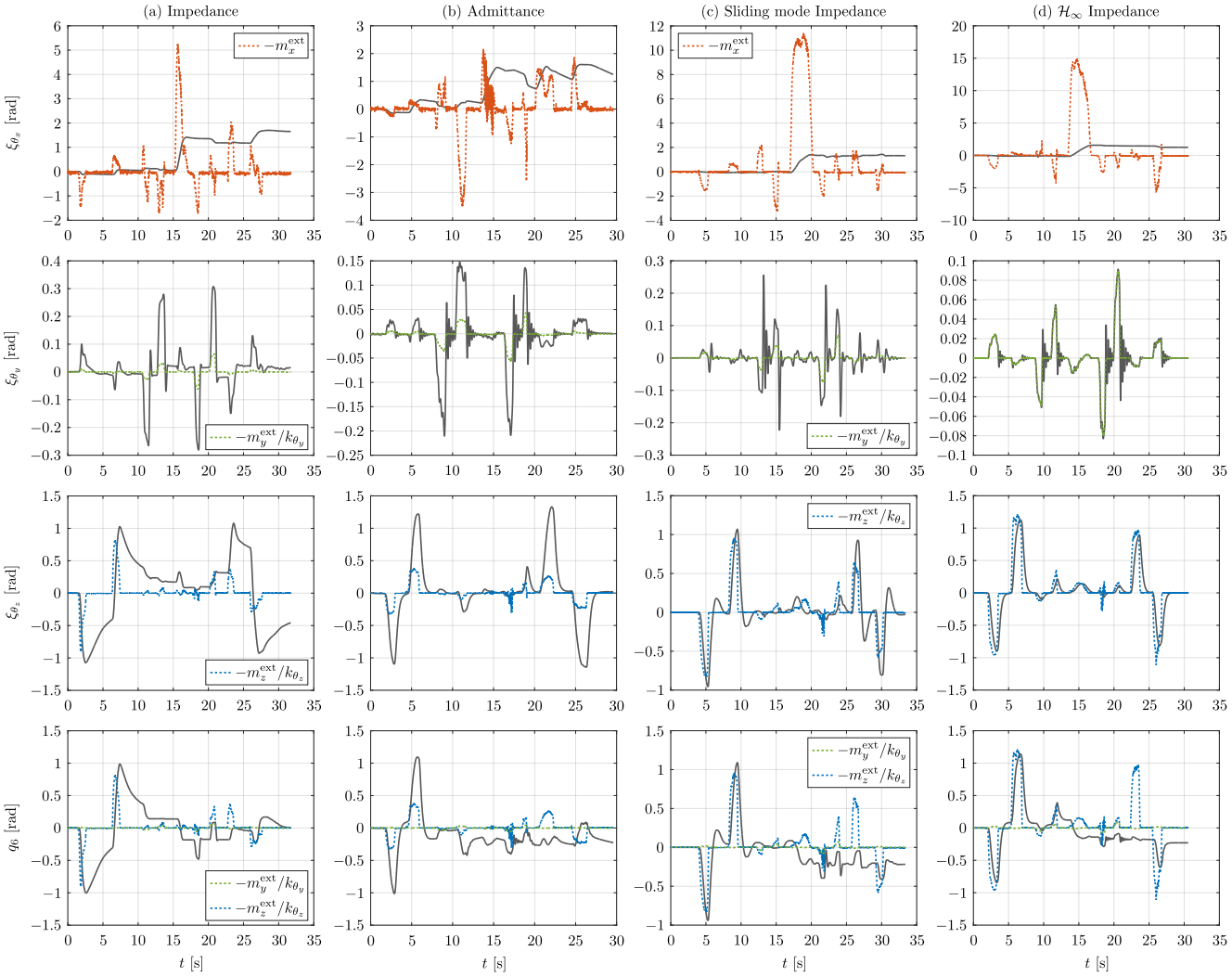


Fig. 11. Experimental results of Experiment D: A representative trial out of the 30 trials.

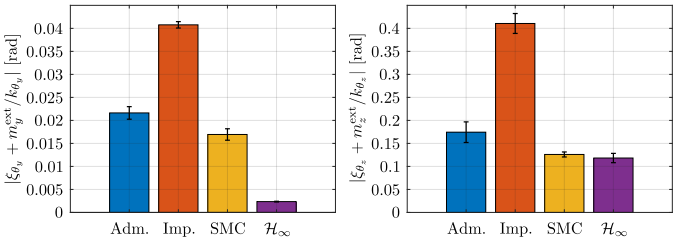


Fig. 12. Statistical analysis of the experimental results of Experiment D.

also responds rapidly and shows superior performance.

E. Application study: grinding task

As an application study (Experiment E), a grinding task was designed to demonstrate the performance of the proposed controller in a highly dynamic scenario. Grinding involves severe fluctuations in the measured force signal—even in the absence of contact—and these fluctuations become more pronounced during contact with a stiff workpiece. Therefore, grinding provides a representative benchmark for evaluating robustness and stability under highly dynamic conditions.

The experimental setup is shown in Fig. 13, where a grinder (GWS 7-100 ET, Bosch) equipped with a diamond cutter (2 608 900 531, Bosch) is mounted on the robot's end-effector, and a stiff workpiece is fixed to the floor within the robot's workspace. The robot was commanded to execute a sinusoidal profile along the x-axis while gradually descending along the z-axis, such that contact occurred approximately 5 s after the start of the motion. The final desired position z_{des} was set approximately 8 mm below the contact surface. The impedance parameters were configured as follows:

$$\begin{aligned}\boldsymbol{\Lambda} &= \mathrm{diag}(10, 10, 10, 1, 1, 1),\\ \boldsymbol{D} &= \mathrm{diag}(400, 400, 400, 40, 40, 40),\\ \boldsymbol{K} &= \mathrm{diag}(1000, 1000, 1000, 400, 400, 400).\end{aligned}$$

In this experiment, the proposed controller was compared with the admittance control with bidirectional energy flow modulation and the sliding mode impedance control. The sliding mode control gains were set as

$$\boldsymbol{K}_1 = \mathrm{diag}(30, 30, 25, 25, 25, 25), \quad \boldsymbol{K}_2 = 0.1\boldsymbol{I}.$$

The experimental results are shown in Figs. 14 and 15.

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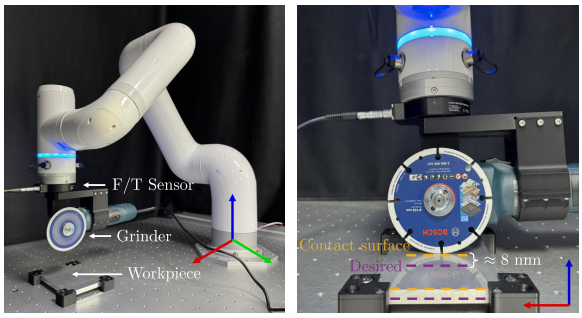


Fig. 13. Experimental setup for Experiment E.

We first examine the results of the admittance control with bidirectional energy flow modulation. Even during the non-contact phase ($t = 0$–5 s), the controller reacts to the severe force signal fluctuations generated by the grinder, causing β to increase rapidly and remain at a high level. As a consequence, the loss of robustness leads to degraded tracking performance even in free space. After contact occurs ($t = 5$–30 s), the modulation parameter β remains close to one, and the controller effectively behaves like the pure PD-based impedance control. As a result, the robot fails to follow the desired trajectory along the x-axis and instead remains confined within a limited region. Along the z-axis, the controller applies excessive contact forces reaching approximately 100–150 N.

In contrast, both the proposed controller and the sliding mode impedance controller maintain robust performance despite the severe force fluctuations, achieving reliable position tracking along the x-axis while keeping the contact force along the z-axis within a moderate range of approximately 40–50 N. Nevertheless, the sliding mode impedance controller exhibits larger sliding error than the proposed controller, with several sharp error spikes during which the robot temporarily loses contact and rebounds before rapidly recovering. In particular, these fluctuations perturb the sliding variable, to which the sliding mode impedance controller responds sensitively, resulting in overshoot. Despite sharing the same sliding surface, the proposed controller maintains stable contact throughout the experiment.

F. Gain tuning and performance variation

As a final experiment (Experiment F), we investigate how the performance of the proposed controller varies with the REF gain $\alpha(\boldsymbol{K}_\gamma + (1/\gamma^2)\boldsymbol{I})$. To this end, we varied α, while γ and $\boldsymbol{K}_\gamma$ were fixed. The following three cases were considered:

$$\text{C1}: \ \alpha = 0.25, \quad \text{C2}: \ \alpha = 0.5, \quad \text{C3}: \ \alpha = 1,$$

where C3 corresponds to the value used in all the previous experiments. To capture the performance variation during both free motion and physical interaction, we revisited Experiment B; all settings other than α were identical to those used therein. The experimental results are presented in Fig. 16.

As shown in Fig. 16, the reference error decreases in inverse proportion to the REF gain. As the REF gain doubles from C1 to C2 and from C2 to C3, the reference error is approximately halved ($0.379 \rightarrow 0.186 \rightarrow 0.095$ m/s). This is consistent

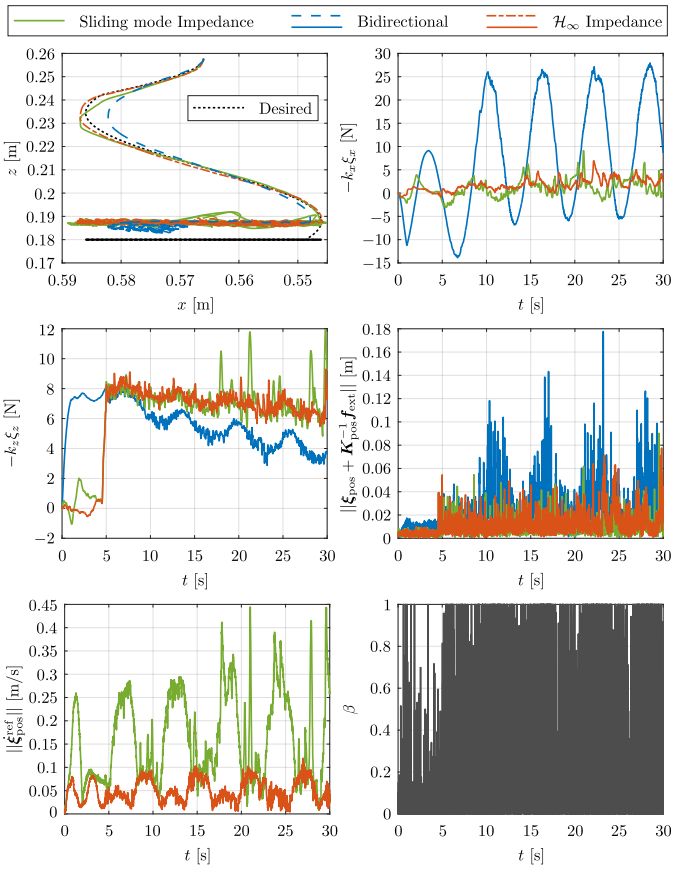


Fig. 14. Experimental results of Experiment E.

with the ISS analysis in Section IV, in which the bound on $\boldsymbol{s}$ is inversely proportional to the REF gain $\alpha(\boldsymbol{K}_\gamma + (1/\gamma^2)\boldsymbol{I})$. From this observation, it can be inferred that, when $(1/\gamma^2)\boldsymbol{I}$ is dominant, i.e., $(1/\gamma^2)\boldsymbol{I} \gg \boldsymbol{K}_\gamma$, decreasing γ improves the performance quadratically, whereas when $\boldsymbol{K}_\gamma$ is dominant, i.e., $\boldsymbol{K}_\gamma \gg (1/\gamma^2)\boldsymbol{I}$, increasing $\boldsymbol{K}_\gamma$ improves the performance linearly. This is consistent with the philosophy of the square-linear tuning rule [18], [19]. This predictable performance variation facilitates gain tuning. In practice, the REF gain can be tuned heuristically by increasing it from a low value until the actuator bandwidth limit is reached.

It can also be observed that the equilibrium position tracking error decreases as the REF gain increases, although not exactly in inverse proportion ($0.00799 \rightarrow 0.00441 \rightarrow 0.00310$ m).

VIII. Conclusions

In this paper, a nonlinear $\mathcal{H}_\infty$ impedance control for robotic systems was proposed. The associated HJI equation was derived for the interconnected robot–environment system and solved analytically. The proposed analytic solution eliminated the artificial gain condition present in previous works. The resulting control law takes the form of RMC with REF. Unlike the existing nonlinear $\mathcal{H}_\infty$ impedance control via the NRIC framework, the proposed controller preserves impedance causality.

Notably, although the environmental dynamics was explicitly incorporated into the control design, it was completely

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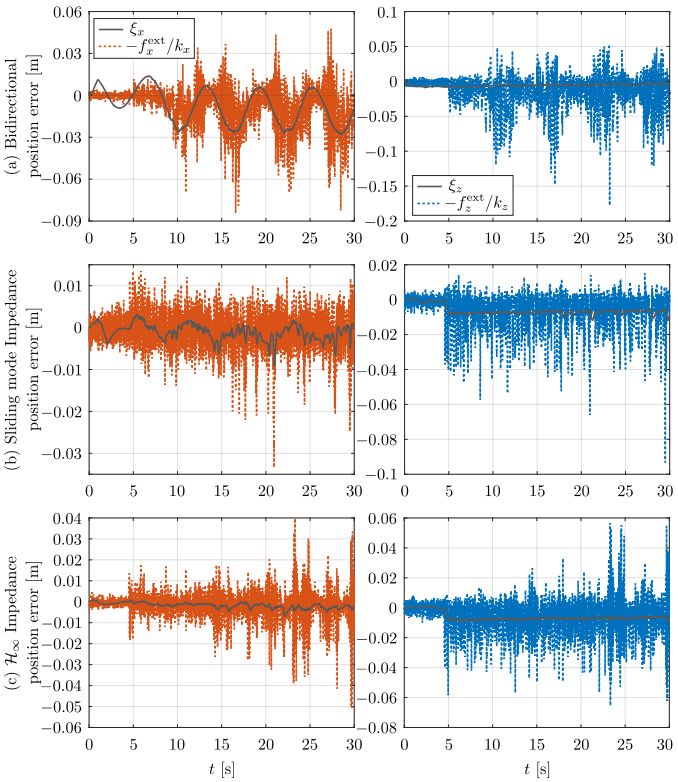


Fig. 15. (Continued) Experimental results of Experiment E.

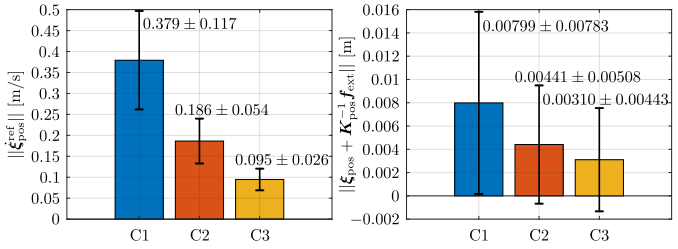


Fig. 16. Experimental results of Experiment F.

canceled out within the resulting control law, enabling generalization to any environment satisfying the mild assumptions. This implies that the nonlinear $\mathcal{H}_\infty$ optimality in impedance control can be achieved without explicit knowledge of the true environmental dynamics. In this sense, the proposed nonlinear $\mathcal{H}_\infty$ impedance control is consistent with the philosophy of the nonlinear $\mathcal{H}_\infty$ position control via RMC with REF.

One potential drawback of the proposed controller is its reliance on force feedback, which makes it sensitive to sensor noise and delay. Nevertheless, owing to its inherent impedance causality, the proposed controller is less affected by these factors than the existing nonlinear $\mathcal{H}_\infty$ impedance control with admittance causality. Another limitation is that the present formulation assumes constant impedance parameters. The extension to variable impedance control, while preserving the nonlinear $\mathcal{H}_\infty$ optimality, is left for future work.

Finally, although the theoretical derivation to rigorously establish the nonlinear $\mathcal{H}_\infty$ optimality was mathematically involved, the resulting control law remains simple, and from a practical point of view, its implementation is straightforward.

APPENDIX
ANALYTIC SOLUTION TO HJI FOR DIAGONAL GAIN CASE

In [17], a class of analytic solutions to the DRE (reduced from the HJI) was presented, where the commutativity of scalar gains k_d and k_p was exploited within each block of the matrix differential equation. For diagonal gains $\boldsymbol{K}_d$ and $\boldsymbol{K}_p$, however, this commutativity no longer holds in general, and thus the applicability of the analytic solution remains unclear. To resolve this issue, this appendix extends the class of analytic solutions to the diagonal gain case.

Following the procedure of [17], we begin by partitioning the $3n \times 3n$ matrices $\boldsymbol{P}(\boldsymbol{x},t)$ and $\boldsymbol{Q}(\boldsymbol{x},t)$ as follows:

$$\boldsymbol{P}(\boldsymbol{x},t) = \begin{bmatrix} \boldsymbol{P}_{11}(\boldsymbol{x},t) & \boldsymbol{P}_{12}(\boldsymbol{x},t) & \boldsymbol{P}_{13}(\boldsymbol{x},t) \\ \boldsymbol{P}_{12}^T(\boldsymbol{x},t) & \boldsymbol{P}_{22}(\boldsymbol{x},t) & \boldsymbol{P}_{23}(\boldsymbol{x},t) \\ \boldsymbol{P}_{13}^T(\boldsymbol{x},t) & \boldsymbol{P}_{23}^T(\boldsymbol{x},t) & \boldsymbol{P}_{33}(\boldsymbol{x},t) \end{bmatrix},$$

$$\boldsymbol{Q}(\boldsymbol{x},t) = \begin{bmatrix} \boldsymbol{Q}_{11}(\boldsymbol{x},t) & \boldsymbol{Q}_{12}(\boldsymbol{x},t) & \boldsymbol{Q}_{13}(\boldsymbol{x},t) \\ \boldsymbol{Q}_{12}^T(\boldsymbol{x},t) & \boldsymbol{Q}_{22}(\boldsymbol{x},t) & \boldsymbol{Q}_{23}(\boldsymbol{x},t) \\ \boldsymbol{Q}_{13}^T(\boldsymbol{x},t) & \boldsymbol{Q}_{23}^T(\boldsymbol{x},t) & \boldsymbol{Q}_{33}(\boldsymbol{x},t) \end{bmatrix},$$

which reduces (10) to a set of six simultaneous $n \times n$ matrix differential equations, composed of three diagonal blocks

$$\begin{aligned} \boldsymbol{0} &= \dot{\boldsymbol{P}}_{11} - \boldsymbol{P}_{13}\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{K}_p - \boldsymbol{K}_p\boldsymbol{C}^T\boldsymbol{M}^{-1}\boldsymbol{P}_{13}^T \\ &\quad - \boldsymbol{P}_{13}\boldsymbol{M}^{-1}\boldsymbol{K}_\gamma\boldsymbol{M}^{-1}\boldsymbol{P}_{13}^T + \boldsymbol{Q}_{11}, \\ \boldsymbol{0} &= \dot{\boldsymbol{P}}_{22} + \boldsymbol{P}_{12} + \boldsymbol{P}_{12}^T - \boldsymbol{P}_{23}\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{K}_d - \boldsymbol{K}_d\boldsymbol{C}^T\boldsymbol{M}^{-1}\boldsymbol{P}_{23}^T \\ &\quad - \boldsymbol{P}_{23}\boldsymbol{K}_p - \boldsymbol{K}_p\boldsymbol{P}_{23}^T - \boldsymbol{P}_{23}\boldsymbol{M}^{-1}\boldsymbol{K}_\gamma\boldsymbol{M}^{-1}\boldsymbol{P}_{23}^T + \boldsymbol{Q}_{22}, \\ \boldsymbol{0} &= \dot{\boldsymbol{P}}_{33} + \boldsymbol{P}_{23} + \boldsymbol{P}_{23}^T - \boldsymbol{P}_{33}\boldsymbol{M}^{-1}\boldsymbol{C} - \boldsymbol{C}^T\boldsymbol{M}^{-1}\boldsymbol{P}_{33} \\ &\quad - \boldsymbol{P}_{33}\boldsymbol{K}_d - \boldsymbol{K}_d\boldsymbol{P}_{33} - \boldsymbol{P}_{33}\boldsymbol{M}^{-1}\boldsymbol{K}_\gamma\boldsymbol{M}^{-1}\boldsymbol{P}_{33} + \boldsymbol{Q}_{33}, \end{aligned} \tag{63}$$

and three off-diagonal blocks

$$\begin{aligned} \boldsymbol{0} &= \dot{\boldsymbol{P}}_{12} + \boldsymbol{P}_{11} - \boldsymbol{P}_{13}\boldsymbol{K}_p - \boldsymbol{P}_{13}\boldsymbol{M}^{-1}\boldsymbol{C}\boldsymbol{K}_d \\ &\quad - \boldsymbol{K}_p\boldsymbol{C}^T\boldsymbol{M}^{-1}\boldsymbol{P}_{23}^T - \boldsymbol{P}_{13}\boldsymbol{M}^{-1}\boldsymbol{K}_\gamma\boldsymbol{M}^{-1}\boldsymbol{P}_{23}^T + \boldsymbol{Q}_{12}, \\ \boldsymbol{0} &= \dot{\boldsymbol{P}}_{13} + \boldsymbol{P}_{12} - \boldsymbol{P}_{13}\boldsymbol{K}_d - \boldsymbol{P}_{13}\boldsymbol{M}^{-1}\boldsymbol{C} - \boldsymbol{K}_p\boldsymbol{C}^T\boldsymbol{M}^{-1}\boldsymbol{P}_{33} \\ &\quad - \boldsymbol{P}_{13}\boldsymbol{M}^{-1}\boldsymbol{K}_\gamma\boldsymbol{M}^{-1}\boldsymbol{P}_{33} + \boldsymbol{Q}_{13}, \\ \boldsymbol{0} &= \dot{\boldsymbol{P}}_{23} + \boldsymbol{P}_{22} - \boldsymbol{P}_{23}\boldsymbol{K}_d + \boldsymbol{P}_{13} - \boldsymbol{K}_p\boldsymbol{P}_{33} - \boldsymbol{P}_{23}\boldsymbol{M}^{-1}\boldsymbol{C} \\ &\quad - \boldsymbol{K}_d\boldsymbol{C}^T\boldsymbol{M}^{-1}\boldsymbol{P}_{33} - \boldsymbol{P}_{23}\boldsymbol{M}^{-1}\boldsymbol{K}_\gamma\boldsymbol{M}^{-1}\boldsymbol{P}_{33} + \boldsymbol{Q}_{23}. \end{aligned} \tag{64}$$

Let us examine the first diagonal block in (63). Setting $\boldsymbol{P}_{13} = \boldsymbol{A}_{13}\boldsymbol{M}$ with a constant diagonal matrix $\boldsymbol{A}_{13}$ yields

$$\boldsymbol{0} = \dot{\boldsymbol{P}}_{11} - \boldsymbol{A}_{13}\boldsymbol{C}\boldsymbol{K}_p - \boldsymbol{K}_p\boldsymbol{C}^T\boldsymbol{A}_{13} - \boldsymbol{A}_{13}^2\boldsymbol{K}_\gamma + \boldsymbol{Q}_{11}. \tag{65}$$

Here, $\boldsymbol{K}_\gamma$ is assumed to be diagonal. If we set $\boldsymbol{P}_{11} = \boldsymbol{A}_{13}\boldsymbol{M}\boldsymbol{K}_p + \overline{\boldsymbol{P}}_{11}$ with a constant symmetric $\overline{\boldsymbol{P}}_{11}$, as in [17], the first three terms on the right-hand side of (65) no longer cancel out since $\boldsymbol{A}_{13}\boldsymbol{C}^T\boldsymbol{K}_p \neq \boldsymbol{K}_p\boldsymbol{C}^T\boldsymbol{A}_{13}$. This remains the case even if the diagonal matrix $\boldsymbol{A}_{13}$ is replaced by a scalar α_{13}. Importantly, this difficulty arises in solving the DRE, not in the HJI itself, which is our primary objective. Since the DRE arises from the HJI via a quadratic form in $\boldsymbol{x}$ [17], each term can be transposed without altering the HJI. Accordingly,

$$\boldsymbol{0} = \boldsymbol{A}_{13}(\dot{\boldsymbol{M}} - 2\boldsymbol{C})\boldsymbol{K}_p - \boldsymbol{A}_{13}^2\boldsymbol{K}_\gamma + \boldsymbol{Q}_{11}. \tag{66}$$

By setting $\boldsymbol{A}_{13} = \alpha\boldsymbol{K}_p$ with a constant scalar α, the first term in (66) is effectively canceled out in the quadratic form of the HJI, which yields $\boldsymbol{Q}_{11} = \alpha^2\boldsymbol{K}_p^2\boldsymbol{K}_\gamma$. In fact, setting

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$\boldsymbol{A}_{13} = \alpha \boldsymbol{K}_p$ directly in (65) yields the exact cancellation of the first three terms, so that the DRE is also satisfied. We now continue along the same line of reasoning.

For the second diagonal block in (63), setting $\boldsymbol{P}_{23} = \boldsymbol{A}_{23}\boldsymbol{M}$ and $\boldsymbol{P}_{22} = \boldsymbol{A}_{23}\boldsymbol{M}\boldsymbol{K}_d + \overline{\boldsymbol{P}}_{22}$ with a constant diagonal $\boldsymbol{A}_{23}$ and a constant symmetric $\overline{\boldsymbol{P}}_{22}$, we have

$$\begin{aligned}\boldsymbol{0} = \boldsymbol{A}_{23}(\dot{\boldsymbol{M}} - 2\boldsymbol{C})\boldsymbol{K}_d + \boldsymbol{P}_{12} + \boldsymbol{P}_{12}^T - 2\boldsymbol{A}_{23}\boldsymbol{M}\boldsymbol{K}_p \\ - \boldsymbol{A}_{23}^2\boldsymbol{K}_\gamma + \boldsymbol{Q}_{22}.\end{aligned}$$

If we set $\boldsymbol{A}_{23} = \alpha \boldsymbol{K}_d$, then the equation reduces further to

$$\boldsymbol{0} = \boldsymbol{P}_{12} + \boldsymbol{P}_{12}^T - 2\alpha \boldsymbol{K}_d \boldsymbol{M} \boldsymbol{K}_p - \alpha^2 \boldsymbol{K}_d^2 \boldsymbol{K}_\gamma + \boldsymbol{Q}_{22}. \quad (67)$$

For the third diagonal block in (63), setting $\boldsymbol{P}_{33} = \alpha \boldsymbol{M}$ and using the property (6), we obtain $\boldsymbol{Q}_{33} = \alpha^2 \boldsymbol{K}_\gamma$.

Upon setting $\boldsymbol{P}_{12} = \alpha \boldsymbol{K}_p \boldsymbol{M} \boldsymbol{K}_d + \overline{\boldsymbol{P}}_{12}$ with a constant symmetric $\overline{\boldsymbol{P}}_{12}$, substituting the above results into the off-diagonal blocks (64), and using the property (6), they reduce to $\overline{\boldsymbol{P}}_{11} = \alpha^2 \boldsymbol{K}_p \boldsymbol{K}_d \boldsymbol{K}_\gamma - \boldsymbol{Q}_{12}$, $\overline{\boldsymbol{P}}_{12} = \alpha^2 \boldsymbol{K}_p \boldsymbol{K}_\gamma - \boldsymbol{Q}_{13}$, and $\overline{\boldsymbol{P}}_{22} = \alpha^2 \boldsymbol{K}_d \boldsymbol{K}_\gamma - \boldsymbol{Q}_{23}$, respectively. The second diagonal block of $\boldsymbol{Q}$ is determined from (67) as

$$\boldsymbol{Q}_{22} = \alpha^2(\boldsymbol{K}_d^2 - 2\boldsymbol{K}_p)\boldsymbol{K}_\gamma + (\boldsymbol{Q}_{13} + \boldsymbol{Q}_{13}^T).$$

Finally, we can fix the off-diagonal blocks of $\boldsymbol{Q}$ to be $\boldsymbol{Q}_{12} = \overline{\boldsymbol{Q}}_{12}$, $\boldsymbol{Q}_{13} = \overline{\boldsymbol{Q}}_{13}$, and $\boldsymbol{Q}_{23} = \overline{\boldsymbol{Q}}_{23}$ for some constant symmetric matrices $\overline{\boldsymbol{Q}}_{(\cdot)}$ since $\overline{\boldsymbol{P}}_{(\cdot)}$ are constant and symmetric. Manipulating the resulting equations, we obtain a class of analytic solutions, as given in (30), which can be verified by direct substitution to satisfy the DRE (10).

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